

The middle and the ends IV.

Endpoint crossings for random walks with memory

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Abstract

Drop a ball through a Galton board on which every bounce after the first repeats the previous direction with probability p . Kagey asked when the middle bin is as likely as each end bin. For this Markov walk with N steps the crossing comes when the expected number of direction changes is about $\log N$. We ask what changes when the ball remembers more than its last step. In the elephant random walk each step copies a uniformly chosen earlier step and reverses it with probability ε . We prove that the crossing is unique and that $N\varepsilon$ at the crossing is $2\log N - \log \log N - \log 16 + o(1)$, about twice the Markov value, because a walk that ends in the middle needs a reversal among its first few steps. At the crossing the law has a hump near each end, and the middle is the lowest bin outside a window of width $O(N^{1/2} \log N)$. For alternating runs whose lengths have tail $k^{-\alpha}$ with $1 < \alpha < 2$, started in equilibrium, the ends beat the middle for large N exactly when α is below the golden ratio. For aging walks, which switch at step k with probability c/k , the crossing with a fixed first step is exactly $c = 1$.

Keywords: persistent random walk; elephant random walk; random recursive tree; Pólya urn; alternating renewal process; heavy tails; golden ratio; Galton board.

MSC2020: 60G50; 60K05; 60C05; 60F05; 60J10.

1 Introduction

Suppose you drop a ball through a Galton board with N rows of pegs. At the first peg the ball goes left or right with probability $\frac{1}{2}$ each. At every later peg it repeats its previous direction with probability p and switches with probability $q = 1 - p$. This is the board of Problem 131 in Kagey’s Open Problem Collection [1], and one of his questions is when the middle bin is exactly as likely as each of the 2 end bins. If p is close to 1 the ball rarely switches, so the ends win. If $p = \frac{1}{2}$ the ball does a simple random walk, so the middle wins. Paper I of this series [2] finds the crossing in between: the middle catches up with the ends when the expected number of switches $T = Nq$ is about $\log N$.

Kagey’s ball remembers 1 step, since its next direction depends only on its last one. In this paper we ask what happens to the middle-versus-ends question if the ball remembers more. We study 3 kinds of memory.

- In the elephant random walk of Schütz and Trimper [5] the ball remembers its whole past. At each peg it picks one of its earlier directions uniformly at random and repeats it, except that with probability ε it reverses it. We write $x = N\varepsilon$, which is the expected number of reversals up to ε .

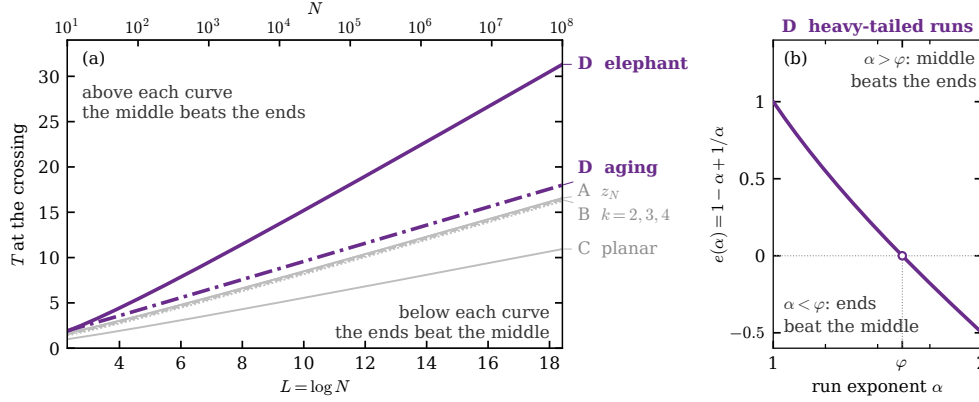


Figure 1: The crossing on each model's switching scale against $L = \log N$ (a), and the heavy-tailed exponent $e(\alpha) = 1 - \alpha + 1/\alpha$ of this paper for a stationary start (b). The scale is T for Papers I to III, $x = N\varepsilon$ for the elephant walk (drawn through the root g_N of Theorem 2.7), and the expected number $H_N - 1$ of switches at the fixed-start crossing $c = 1$ for the aging walk. The curves of this paper are in bold.

- In a walk with heavy-tailed runs the ball remembers how long it has gone in its current direction. Its path is a sequence of runs of alternating direction, and the run lengths ξ are independent with $\mathbb{P}(\xi \geq k) = k^{-\alpha}$ for $k \geq 1$.
- In an aging walk the ball remembers how old it is. It switches at peg $k \geq 2$ with probability c/k .

The answer depends on the kind of memory. For the elephant walk the middle needs about twice as many reversals as Kagey's board needs switches: the crossing value of x is about $2 \log N$, not $\log N$. For heavy-tailed runs started in equilibrium the ends win for large N if α is below the golden ratio $\varphi = (1 + \sqrt{5})/2$, and the middle wins if α is above φ . For the aging walk with a fixed first step the crossing is exactly $c = 1$, for every N .

Why the elephant needs twice as many reversals. The ends are easy. The ball lands in the right end bin exactly when its first step goes right and it never changes direction. This has probability $\frac{1}{2}(1 - \varepsilon)^{N-1} \approx \frac{1}{2}e^{-x}$ for the elephant walk, and the same with q in place of ε for Kagey's board. So only the middle differs. On Kagey's board the T switches are spread over the whole walk, and each moves the final position by about a run length N/T . So the number of right steps spreads over about $N/\sqrt{T}$ bins, the middle bin has probability of order $\sqrt{T}/N$, and the ends lose when e^{-T} falls to about $1/N$, that is, when $T \approx \log N$.

The elephant walk spreads a reversal in a different way. A reversal at step $t + 1$ is copied by later steps, those copies are copied, and so on. The steps that descend from it form a family, whose share of the walk behaves like the share of 1 ball in Pólya's urn started with 1 ball against t . So a late reversal has a small family and hardly moves the ball. To land exactly in the middle, the walk needs a family that fills half the walk, so it needs a reversal among its first few steps. This costs a factor ε , and hitting the exact middle costs another factor of order $1/N$. So the middle bin has probability of order $\varepsilon/N = x/N^2$ (in fact about $4x/N^2$), and the ends lose only when e^{-x} falls to about $1/N^2$, that is, when $x \approx 2 \log N$. More precisely, at Kagey's crossing $\frac{1}{2}e^{-T} \approx \sqrt{2T/\pi}/N$, and squaring this relation turns it into the elephant relation $\frac{1}{2}e^{-x} \approx 4x/N^2$ with $x \approx 2T - \log(2\pi)$.

The small families also give the shape of the law. About x/m^2 reversals have a family of m

steps, so the bin at distance m from an end has probability about $x/(2m^2)$ over a wide range of m . So the law has a hump near each end, and the middle is where the 2 tails meet: twice $x/(2m^2)$ at $m = N/2$ is the $4x/N^2$ above.

Why the golden ratio. For heavy-tailed runs take $1 < \alpha < 2$, so a run has a finite mean μ but an infinite variance. Start the walk in equilibrium, at a typical time of a long sequence of runs. Then the first run is already in progress, and long runs are more likely to be in progress. The chance that this run covers all N steps is of order $N^{1-\alpha}$, and this is the end. For the middle, the runs to the right and the runs to the left form 2 independent renewal processes, each making about $N/(2\mu)$ runs, and the ball ends in the middle when they meet at level $N/2$. Each process fluctuates on the scale $N^{1/\alpha}$ of a stable law, so they meet exactly with probability of order $N^{-1/\alpha}$. Hence the ratio of end to middle is of order $N^{1-\alpha}/N^{-1/\alpha} = N^{1-\alpha+1/\alpha}$. The exponent vanishes when $\alpha^2 - \alpha - 1 = 0$, that is, at $\alpha = \varphi$. Below φ the ends win, and above φ the middle wins. If the first run is a fresh run instead, the end is of order $N^{-\alpha}$, a factor N smaller, and the middle wins for every $1 < \alpha < 2$.

1.1 The series

This is Paper IV of the series *The middle and the ends*. Paper I [2] solves Kagey's board, Paper II [3] treats d directions on the complete graph, Paper III [4] gives the general theory for slowly switching chains, and Paper IV asks what memory changes. The 4 papers use the same notation.

- Kagey's board: N steps (Kagey's row n has $N = n - 1$). The first step is left or right with probability $\frac{1}{2}$, and each later step repeats the previous one with probability p . We put $q = 1 - p$ and $u = q/p$ (the odds of a switch).
- $P_N(k)$: the probability of k right steps, that is, that the ball lands in bin k . The endpoints have $P_N(0) = P_N(N) = p^{N-1}/2$.
- $L = \log N$, and $T = Nq$ is the expected number of switches (up to the first step). The crossing is p_N , and $z_N = N(1 - p_N)$.
- "The crossing": the value of p (or T) where a bin is exactly as likely as an endpoint.
- Multistate walks (Papers II and III): d states, a "face" F is a nonempty set of states, and the "vertices" are the faces of size 1. For a chain with transition matrix $I + (T/N)Q$, λ_F is the principal Dirichlet eigenvalue of $-Q$ on F , and K_F is a second-order constant.

Papers I to III study Markov walks, whose next step depends only on the current one, and their answer is that the middle (or a face) ties with the vertices when T is of order $\log N$. This paper keeps Kagey's question and changes the walk. For the elephant walk $x = N\varepsilon$ plays the role of T and its crossing x_N plays the role of z_N . We compare the 2 through the crossing equation of Paper I, which says that for every $N \geq 2$

$$z_N + \frac{1}{2} \log z_N = L + c_0 + \frac{1}{8z_N} + O(L^{-2}), \quad c_0 = \frac{1}{2} \log \frac{\pi}{8}, \quad (1)$$

where the error term is at most $\frac{1}{4L^2} + \frac{L^2+L}{N}$ in absolute value, and that $L/2 \leq z_N \leq L$. As in Paper II the start law matters: for heavy-tailed runs it decides whether the threshold is φ or 1. Figure 1 shows the crossings of all 4 papers.

1.2 Setting and notation

Every walk in this paper takes N steps $X_1, \dots, X_N \in \{+1, -1\}$, where $+1$ means right. Let $A_N = \#\{i \leq N : X_i = +1\}$ be the number of right steps, and keep writing $P_N(k) = \mathbb{P}(A_N = k)$

for the probability that the ball lands in bin k , now for whichever walk we study. Unless we say otherwise the first step is fair, $\mathbb{P}(X_1 = +1) = \frac{1}{2}$, and the rules treat the 2 directions alike, so $P_N(k) = P_N(N - k)$. We write $\mathbb{P}_+$ and $\mathbb{E}_+$ for probability and expectation given $X_1 = +1$. For even N we put $h = N/2$ and call $C_N = P_N(h)$ the center and $E_N = P_N(N) = P_N(0)$ the end. Whenever C_N appears, N is even. The crossing is the value of the parameter at which $C_N = E_N$. All logarithms are natural, $L = \log N$, $H_k = \sum_{j \leq k} 1/j$ is the k -th harmonic number, and γ is Euler's constant. Limits, $O(\cdot)$ and $o(\cdot)$ refer to $N \rightarrow \infty$, over even N when C_N appears. The constants implied by $O(\cdot)$ are absolute unless we name what they depend on.

The elephant walk (Definition 2.1) has the reversal probability $\varepsilon \in (0, 1)$ and $x = N\varepsilon$. Its crossing is ε_N , and $x_N = N\varepsilon_N$. Heavy-tailed runs (Section 4) have the exponent α , the mean run length $\mu = \mathbb{E}\xi = \zeta(\alpha)$ for $\alpha > 1$, and the length D of the first run. A fresh start has D distributed as ξ , and the stationary start has $\mathbb{P}(D = d) = \mathbb{P}(\xi \geq d)/\mu$. We put $R_N = E_N/C_N$, so the ends win exactly when $R_N > 1$. The aging walk (Remark 4.8) has the constant $c \in (0, 2)$. The symbols that only one section uses are defined at the start of that section. The letters p , q and u have their meaning for Kagey's board only in this introduction, and later sections reuse them locally.

1.3 Results

The elephant walk. For every even $N \geq 2$ the crossing ε_N exists, is unique and is below $\frac{1}{2}$ (Proposition 2.3). The center has derivative exactly $4/(N+2)$ at $\varepsilon = 0$, and $C_N = \frac{4\varepsilon}{N+2}(1 + O(\varepsilon \log N))$ for even $N \geq 64$ and $\varepsilon(1 + \log N) \leq \frac{1}{100}$, with explicit constants (Proposition 2.5 and Theorem 2.6). This gives $x_N + \log x_N = 2L - \log 8 + O(L^2/N)$ (Theorem 2.7) and $x_N = 2z_N - \log(2\pi) + c_*/L + O(\log L/L^2)$ with $c_* = \frac{1}{2} \log(2\pi) - \frac{1}{4}$ (Theorem 2.8). At the crossing the end is below its neighbor for every even $N \geq 8$ (Corollary 3.6), bin $N - m$ has probability $\frac{x}{2m^2}(1 + o(1))$ for $x \log^2 x \ll m \ll N$ (Theorem 3.8), and for large N the center is below every interior bin outside a window of width of order $N^{1/2} \log N$ (Theorem 3.9), while the modes lie within $O(x_N \log^2 x_N)$ of the ends (Corollary 3.11).

Heavy-tailed runs and aging. For $1 < \alpha < 2$ and any first run, C_N is an explicit constant times $N^{-1/\alpha}$ (Theorem 4.4). With the stationary start $R_N \sim C(\alpha)N^{1-\alpha+1/\alpha}$, and $C(\varphi) = 0.7761\dots$, so at $\alpha = \varphi$ the center still wins. With a fresh start $R_N \rightarrow 0$ (Corollary 4.5). The aging walk is in Remark 4.8. Appendices A to C hold the longer proofs.

1.4 Prior work

The steps of the elephant walk split into clusters, since a step either copies an earlier step exactly, with probability $1 - 2\varepsilon$, or starts a new cluster with a fair sign. This is Kürsten's coupling with bond percolation on random recursive trees [6], and Baur and Bertoin [7] relate the walk to Pólya-type urns. Our regime $N\varepsilon \asymp \log N$ is the strongly supercritical regime of Baur [8] and Bertoin [9], whose results describe cluster sizes, not the center probability, which is a large deviation of the root cluster. Guérin, Laulin, Raschel and Simon [10, proof of Theorem 1.1] prove that the law with a fixed first step is unimodal, and their other results fix the memory parameter as N grows. We did not find an asymptotic formula for the center probability when $\varepsilon \rightarrow 0$ as N grows.

For heavy-tailed runs, Godrèche and Luck [11] give the transform of the occupation time of an alternating renewal process and the scalings of the center and of both edges, and [13, 14] describe the profile of a Lévy walk. Wang, Li and Hänggi [15] already observe, as a heuristic for heat conduction, that equal decay of the central peak and the side peaks gives the golden ratio. We

prove the lattice statement with a local limit at the center, keep the 2 starts apart, and compute $C(\alpha)$. For aging walks, Dietz and Sethuraman [22] proved the limit law of A_N/N , Engländer and Volkov [23] gave a second proof, and Engländer, Volkov and Wang [24] found the scaling limit of the walk.

2 The elephant walk and its crossing

We reduce the elephant walk to a counting chain, prove that the crossing is unique, compute the center to first and second order in ε , and locate the crossing.

Definition 2.1 (elephant walk). Fix $0 < \varepsilon < 1$. Let X_1 be uniform on $\{\pm 1\}$. For $t \geq 1$ let U_t be uniform on $\{1, \dots, t\}$, and put $X_{t+1} = X_{U_t}$ with probability $1 - \varepsilon$ and $X_{t+1} = -X_{U_t}$ with probability ε . All these choices are independent. We call the event $X_{t+1} = -X_{U_t}$ a flip. As in Section 1.2, $A_N = \#\{i \leq N : X_i = +1\}$ and $P_N(k) = \mathbb{P}(A_N = k)$. The crossing ε_N is the value of ε with $C_N = E_N$, and $x_N = N\varepsilon_N$.

A flip is what the introduction called a reversal. In the notation of [5] the memory parameter is $1 - \varepsilon$, and [10] write $a = 1 - 2\varepsilon$. Given the first step, all that matters is how many steps disagree with it. So let M_t be the number of $i \leq t$ with $X_i \neq X_1$, and put

$$r_t(b) = \varepsilon + (1 - 2\varepsilon)\frac{b}{t}, \quad f_N(m) = \mathbb{P}_+(M_N = m).$$

Lemma 2.2. *The process $(M_t)_{t \geq 1}$ is a Markov chain with $M_1 = 0$ and $\mathbb{P}(M_{t+1} = b + 1 \mid M_t = b) = r_t(b) = 1 - \mathbb{P}(M_{t+1} = b \mid M_t = b)$, and it is independent of X_1 . For $0 \leq k \leq N$, $P_N(k) = \frac{1}{2}[f_N(N - k) + f_N(k)]$. In particular $C_N = f_N(h)$, $E_N = \frac{1}{2}(1 - \varepsilon)^{N-1}$, and $P_N(k) = P_N(N - k)$.*

Proof. Given $X_1, \dots, X_t$, the copied step X_{U_t} differs from X_1 with probability M_t/t . So $X_{t+1} \neq X_1$ with conditional probability $(1 - \varepsilon)M_t/t + \varepsilon(1 - M_t/t) = r_t(M_t)$. This depends only on M_t and not on X_1 , which proves the first claim. On $\{X_1 = +1\}$ we have $A_N = N - M_N$, and on $\{X_1 = -1\}$ we have $A_N = M_N$. Averaging over X_1 gives the formula for $P_N(k)$, which does not change under $k \mapsto N - k$. Finally $M_N \leq N - 1$, so $f_N(N) = 0$, and $f_N(0) = \prod_{t=1}^{N-1} (1 - r_t(0)) = (1 - \varepsilon)^{N-1}$. $\square$

So the end is Kagey's end $p^{N-1}/2$ with $p = 1 - \varepsilon$. More flips always help the middle against the end:

Proposition 2.3. *Let $N \geq 2$ and $1 \leq k \leq N - 1$. Then $f_N(k)/f_N(0)$ and $P_N(k)/E_N$ are strictly increasing in $\varepsilon \in (0, 1)$. In particular C_N/E_N is strictly increasing, the crossing ε_N exists and is unique, and $\varepsilon_N < \frac{1}{2}$. Also $C_N \geq 2E_N$ for $\varepsilon \geq \frac{1}{2}$.*

Proof. The probability of a path of M is a product of $N - 1$ factors, namely $r_s(b_s)$ for an up step at time s and $1 - r_s(b_s)$ otherwise, where b_s is the state at time s . We divide it by the probability $f_N(0) = (1 - \varepsilon)^{N-1}$ of the path that never flips. With $y = b_s/s \in [0, 1)$ the factors become $\frac{\varepsilon + (1 - 2\varepsilon)y}{1 - \varepsilon}$ and $1 - y\frac{1 - 2\varepsilon}{1 - \varepsilon}$, whose derivatives in ε are $\frac{1 - y}{(1 - \varepsilon)^2} > 0$ and $\frac{y}{(1 - \varepsilon)^2} \geq 0$. Both factors are positive, since $r_s(b_s) = \varepsilon(1 - y) + (1 - \varepsilon)y$ lies strictly between 0 and 1. So the ratio for each path is nondecreasing in ε , and strictly increasing if the path has an up step. Summing over the paths with $k \geq 1$ up steps proves the claim for $f_N(k)/f_N(0)$, and the claim for $P_N(k)/E_N$ follows by Lemma 2.2. As $\varepsilon \rightarrow 0$, $C_N \rightarrow 0$ and $E_N \rightarrow \frac{1}{2}$. For $\varepsilon \geq \frac{1}{2}$ every $r_s(b)$ and every $1 - r_s(b)$ lies in $[1 - \varepsilon, \varepsilon]$, so each path of M that ends at h has probability at least $(1 - \varepsilon)^{N-1} = 2E_N$. Hence $C_N \geq 2E_N$. The ratio C_N/E_N is continuous and strictly increasing, so it equals 1 at exactly one ε , and this ε lies in $(0, \frac{1}{2})$. $\square$

2.1 First order: one early flip

When $\varepsilon = 0$ the chain M never leaves 0. After a first flip, and to first order in ε , it is Pólya's urn.

Lemma 2.4 (Pólya's urn). *Let $2 \leq v \leq N$. At time v an urn holds 1 ball of one color and $v - 1$ of the other. At each time $k = v, \dots, N - 1$ we draw a ball uniformly and return it with 1 more ball of its color. Let $\beta_v(d)$ be the probability that at time N there are d balls of the first color. Then $\beta_v(d) = \binom{N-d-1}{v-2} / \binom{N-1}{v-1}$ for $1 \leq d \leq N - v + 1$, and $\beta_v(d) = 0$ for other d . Also β_v is nonincreasing in d and $\sum_d d \beta_v(d) = N/v$. For $1 \leq d \leq N - 1$,*

$$\sum_{v=2}^N \beta_v(d) = \frac{N}{d(d+1)}, \quad \sum_{v=2}^N (v-2) \beta_v(d) = \frac{2N(N-d-1)}{d(d+1)(d+2)}.$$

Proof. If the urn holds s balls of the first color at time k , the next draw is of that color with probability s/k . So a given order of i draws of the first color among the $N - v$ draws has probability $\frac{i!(v-1)v \dots (N-i-2)}{v(v+1) \dots (N-1)} = \frac{(v-1)! i! (N-i-2)!}{(N-1)!}$. Multiplying by $\binom{N-v}{i}$ and putting $d = i + 1$ gives $\beta_v(d) = a_d(v-1) \binom{N-v}{d-1}$ with $a_d = \frac{(d-1)!(N-d-1)!}{(N-1)!}$, which equals $\binom{N-d-1}{v-2} / \binom{N-1}{v-1}$. Hence $\beta_v(d)/\beta_v(d-1) = 1 - \frac{v-2}{N-d}$ for $2 \leq d \leq N - v + 1$, and β_v is nonincreasing. The fraction of balls of the first color is a martingale that starts at $1/v$, which gives the mean. For the 2 sums put $i = N - v$, so that $v - 1 = \binom{N-1-i}{1}$ and $(v-1)(v-2) = 2 \binom{N-1-i}{2}$. By the Chu–Vandermonde identity, $\sum_i \binom{i}{d-1} \binom{N-1-i}{1} = \binom{N}{d+1}$ and $\sum_i \binom{i}{d-1} \binom{N-1-i}{2} = \binom{N}{d+2}$. Multiplying by a_d and $2a_d$ gives the 2 sums. $\square$

We need 3 more pieces of notation. Let $V_s(b) = \mathbb{P}_+(M_N = h \mid M_s = b)$ be the chance of ending at the center from state b at time s . For $1 \leq t \leq h$ let

$$\pi_t = \beta_{t+1}(h) = \frac{t}{N-t} \prod_{i=1}^{t-1} \frac{h-i}{N-i}, \quad c_1 = \sum_{t=1}^h \pi_t, \quad w_t = \frac{\pi_t}{c_1},$$

so that $(w_t)_{1 \leq t \leq h}$ is a probability vector.

Proposition 2.5. *Let $N \geq 2$ be even.*

(a) $C_N = \sum_{t=1}^h \varepsilon(1-\varepsilon)^{t-1} V_{t+1}(1).$

(b) C_N is a polynomial in ε with $C_N(0) = 0$. At $\varepsilon = 0$ we have $V_{t+1}(1) = \pi_t$, and $C'_N(0) = c_1 = \frac{4}{N+2}$ exactly.

(c) $\sum_t w_t t = 3 - \frac{6}{h+2}$, and $w_t \leq 2t2^{-t}$ for $1 \leq t \leq h$.

Proof. (a) Under $\mathbb{P}_+$ the first flip happens at step $t + 1$ exactly when $M_s = 0$ for $s \leq t$ and $M_{t+1} = 1$. Since $r_s(0) = \varepsilon$, this has probability $(1-\varepsilon)^{t-1}\varepsilon$, and by the Markov property the chain then ends at h with probability $V_{t+1}(1)$. At time $t + 1$ there are t steps equal to X_1 , and this number never decreases. So $V_{t+1}(1) = 0$ for $t > h$. Paths with no flip end at $0 \neq h$.

(b) Every transition probability is affine in ε , so C_N and $V_{t+1}(1)$ are polynomials in ε . By (a), $C_N(0) = 0$ and $C'_N(0) = \sum_t V_{t+1}(1)|_{\varepsilon=0}$. At $\varepsilon = 0$ we have $r_s(b) = b/s$. So from time $t + 1$ on, M counts the balls of the first color in the urn of Lemma 2.4 with $v = t + 1$, and $V_{t+1}(1) = \beta_{t+1}(h) = \pi_t$. Since $\beta_v(h) = 0$ for $v > h + 1$, Lemma 2.4 with $d = h$ gives $c_1 = \sum_{v=2}^N \beta_v(h) = \frac{N}{h(h+1)} = \frac{4}{N+2}$.

(c) In the same way the second sum of Lemma 2.4 gives $\sum_t (t-1)\pi_t = \frac{2N(h-1)}{h(h+1)(h+2)}$, so $\sum_t w_t(t-1) = \frac{2(h-1)}{h+2}$. Next $\frac{h-i}{N-i} \leq \frac{1}{2}$, so $w_t = \frac{N+2}{4} \pi_t \leq \frac{N+2}{2(N-t)} t 2^{-t} \leq 2t2^{-t}$ for $t \leq h$. $\square$

N	x_N	g_N	$x_N - x_{2\text{nd}}$	$x_N / \log N$	$D(N) \log N$
50	3.940487	4.288636	+0.3452	1.007	-0.083
400	7.629217	7.843768	+0.0621	1.273	-0.199
1600	10.240565	10.340051	+0.0179	1.388	+0.215
3200	11.546834	11.610464	+0.0097	1.431	+0.386
12800	14.158845	14.182921	+0.0028	1.497	+0.607
51200	16.778468	16.786946	+0.0008	1.547	+0.707

Table 1: Exact elephant crossings, the root g_N of $G(g) = \Lambda_N$, and the difference $D(N)$ of Theorem 2.8 times $\log N$. Here $x_{2\text{nd}}$ is the second-order formula of Remark 2.9 without its error term.

In words, to first order the walk flips once, at some step $t + 1$, and then runs as Pólya's urn from t balls of one color and 1 of the other. The minority fraction is then close to a Beta(1, t) variable, whose density at $\frac{1}{2}$ is $t2^{1-t}$, and $\sum_t t2^{1-t} = 4$ explains the constant $4/(N + 2)$. Since $w_t \leq 2t2^{-t}$, only an early flip counts.

2.2 Second order

Theorem 2.6. *Let $N \geq 2$ be even.*

(a) *For $0 < \varepsilon < \frac{1}{2}$, $C_N \geq \frac{4\varepsilon}{N+2}(1 - 2\varepsilon)$.*

(b) *$C_N \leq \frac{4\varepsilon}{N+2} N^{2\varepsilon} e^{5200\varepsilon}$ whenever $N \geq 64$ and $0 < \varepsilon(1 + \log N) \leq \frac{1}{100}$.*

In particular $\log C_N = \log \frac{4\varepsilon}{N+2} + O(\varepsilon \log N)$ in the range of (b), with an absolute implied constant.

In words, the first-order term is correct up to a factor $1 + O(\varepsilon \log N)$, which is $1 + O(L^2/N)$ at the crossing.

Idea of the proof. The proof is in Appendix A. After the first flip, M at time s is Pólya's urn with βs extra balls of each color, where $\beta = \varepsilon/(1 - 2\varepsilon)$ (Lemma A.1). They pull the minority fraction toward $\frac{1}{2}$. For (a) we drop them. For (b) we compare M with an urn with $\kappa(s + 10)$ extra balls, $\kappa = 2\varepsilon/(1 - 4\varepsilon)$; the step at time s costs a factor of about $e^{\kappa/s}$, and the product is about $e^{\kappa H_N} \approx N^{2\varepsilon}$.

2.3 The crossing

By Theorem 2.6, $C_N \approx 4x/N^2$ when $\varepsilon = x/N$ and x is of order $\log N$. Also $E_N = \frac{1}{2}(1 - x/N)^{N-1} \approx \frac{1}{2}e^{-x}$. Setting them equal gives $x + \log x \approx 2 \log N - \log 8$. So let

$$\Lambda_N = 2L - \log 8, \quad G(x) = x + \log x,$$

and let g_N be the root of $G(g_N) = \Lambda_N$. It exists and is unique for $\Lambda_N > 1$, since G increases from $-\infty$ to ∞ on $(0, \infty)$.

Theorem 2.7. *As $N \rightarrow \infty$ over even N , the following hold.*

(a) *$x_N + \log x_N = \Lambda_N + O(L^2/N)$, so $x_N = g_N + O(L^2/N)$.*

(b) *$x_N = 2 \log N - \log \log N - \log 16 + O(\log \log N / \log N)$.*

In words, the root g_N locates the crossing to within $O(L^2/N)$, and the expected number of flips at the crossing is about $2 \log N$.

Proof. (a) *Step 1: both sides near the crossing.* Put $F(x) = \log C_N(x/N) - \log E_N(x/N)$ for $0 < x < N$, where $C_N(\varepsilon)$ and $E_N(\varepsilon)$ are the center and the end at flip probability ε . By Proposition 2.3, F is strictly increasing and x_N is its only zero. Let $1 \leq x \leq 2\Lambda_N$. Then $x/N \leq 4L/N$, so for large N Theorem 2.6 applies with $\varepsilon = x/N$. As $\log \frac{4\varepsilon}{N+2} = \log x + \log 4 - 2L - \log(1 + 2/N)$ and $x \geq 1$, it gives $\log C_N = \log x + \log 4 - 2L + O(xL/N)$. Also $\log(1 - y) = -y + O(y^2)$ for $0 \leq y \leq \frac{1}{2}$, so $\log E_N = -\log 2 - (N - 1)x/N + O(x^2/N) = -\log 2 - x + O(x^2/N)$. Since $\log 4 + \log 2 = \log 8$ and $x \leq 2\Lambda_N \leq 4L$, we get $|F(x) - G(x) + \Lambda_N| \leq \rho_N$ on $[1, 2\Lambda_N]$, with $\rho_N = O(L^2/N)$.

Step 2: trap the zero. Now $G' \geq 1$ and $G(g_N) = \Lambda_N$. So $F(x) < 0$ for $1 \leq x < g_N - \rho_N$ and $F(x) > 0$ for $g_N + \rho_N < x \leq 2\Lambda_N$. As $\Lambda_N - \log \Lambda_N \leq g_N \leq \Lambda_N$ (because $G(\Lambda_N - \log \Lambda_N) \leq \Lambda_N \leq G(\Lambda_N)$), both ranges are nonempty for large N . Since F is increasing, $|x_N - g_N| \leq \rho_N$. Then $F(x_N) = 0$ gives $|G(x_N) - \Lambda_N| \leq \rho_N$.

(b) Since $g_N = \Lambda_N - \log g_N$ and $\Lambda_N - \log \Lambda_N \leq g_N \leq \Lambda_N$, we have $\log g_N = \log \Lambda_N + O(\log \Lambda_N / \Lambda_N)$, and $\log \Lambda_N = \log 2 + \log \log N + O(1/\log N)$. So $g_N = 2 \log N - \log 8 - \log 2 - \log \log N + O(\log \log N / \log N)$, which is (b) for g_N , and (a) gives it for x_N . $\square$

Table 1 gives exact crossings on the grid $N = 50 \cdot 2^k$, computed from the recursion for f_N in Lemma 2.2. The limit 2 of $x_N / \log N$ is approached slowly, because of the term $-\log \log N$.

Now we compare with Kagey's board. Doubling (1) without its last 2 terms gives $2z_N + \log(2z_N) = \Lambda_N + \log(2\pi)$, so $2z_N - \log(2\pi)$ nearly solves $G(x) = \Lambda_N$. What is left is of order $1/L$.

Theorem 2.8. *Let z_N be the crossing value of T for Kagey's board with N steps, as in Section 1.1. As $N \rightarrow \infty$ over even N ,*

$$D(N) := x_N - (2z_N - \log(2\pi)) = \frac{c_*}{\log N} + O\left(\frac{\log \log N}{(\log N)^2}\right), \quad c_* = \frac{1}{2} \log(2\pi) - \frac{1}{4} = 0.668939 \dots$$

In words, the elephant needs twice as many flips as Kagey's board needs switches, less $\log(2\pi) = 1.84$.

Proof. Write $\ell = \log(2\pi)$. By Section 1.1, the error term in (1) is $O(L^{-2})$ and $L/2 \leq z_N \leq L$. So (1) gives $z_N + \frac{1}{2} \log z_N = L + O(1)$ with $\log z_N = \log L + O(1)$, hence $z_N = L - \frac{1}{2} \log L + O(1)$. Since $z_N \geq L/2$, this gives $\frac{1}{8z_N} - \frac{1}{8L} = \frac{L - z_N}{8Lz_N} = O(\log L / L^2)$. Put $Y = 2z_N$. Twice (1) plus $\log 2$, together with $2c_0 + \log 2 = \log \frac{\pi}{4}$, gives $Y + \log Y = 2L + \log \frac{\pi}{4} + \frac{1}{4L} + O(\log L / L^2)$. Also $Y = 2L - \log L + O(1)$, so $1/Y = 1/(2L) + O(\log L / L^2)$ and $\log(1 - \ell/Y) = -\ell/(2L) + O(\log L / L^2)$. Using $\log \frac{\pi}{4} - \ell = -\log 8$, we get

$$G(Y - \ell) = Y + \log Y - \ell + \log\left(1 - \frac{\ell}{Y}\right) = \Lambda_N + \frac{1}{4L} - \frac{\ell}{2L} + O\left(\frac{\log L}{L^2}\right) = \Lambda_N - \frac{c_*}{L} + O\left(\frac{\log L}{L^2}\right).$$

By Theorem 2.7(a), $G(x_N) = \Lambda_N + O(L^2/N)$. Both x_N and $Y - \ell = 2z_N - \log(2\pi)$ are $2L + O(\log L)$, so $G' = 1 + 1/x = 1 + O(1/L)$ between them, and the mean value theorem gives $x_N - (Y - \ell) = c_*/L + O(\log L / L^2)$. $\square$

We computed z_N from the exact law of Kagey's board [2]. The difference $D(N)$ changes sign between $N = 400$ and $N = 800$, and $D(N) \log N$ is 0.707 at $N = 51200$ against the limit $c_* = 0.669$ (Table 1). So the asymptotic regime is reached slowly.

Remark 2.9 (second order, proofs omitted). Write $V_{t+1}(1)/\pi_t$ as the mean of a product of factors $1 + O(\varepsilon)$ along Pólya's urn conditioned to end at h . Jensen's inequality then gives a lower bound that matches Theorem 2.6(b), so $\log C_N = \log \frac{4\varepsilon}{N+2} + 2\varepsilon \log N + O(\varepsilon)$ in the range of that theorem. With this the proof of Theorem 2.7 gives $x_N = g_N - \frac{2g_N \log N + g_N^2/2}{N(1+1/g_N)} + O(\frac{\log N}{N})$, so $x_N < g_N$ for large N . Write $C_N = c_1\varepsilon + c_2\varepsilon^2 + O(\varepsilon^3)$ and $k_2(N) = c_2/c_1$. Then $k_2(N) - 2\log N$ tends to $2\gamma - 2 - \log 2$, which is -1.5387 to 4 decimals. We omit the details of these 3 results. We do not know an upper bound $\log C_N \leq \log(c_1\varepsilon) + \varepsilon k_2(N) + O(\varepsilon^2(\log N)^k)$, or an explicit N beyond which $g_N > x_N$.

3 The elephant walk near its crossing

We now look at the whole law P_N near the crossing (Figure 2). The picture is that of the introduction: each flip turns a whole family of steps, most families are small, and the law falls like $x/(2m^2)$ from each end. Lemmas 3.1 to 3.4 and Theorem 3.5 hold as stated. Elsewhere in this section and in Appendix B we assume, unless we say otherwise, that

$$N \geq 64, \quad 0 < \varepsilon \leq \frac{1}{4}, \quad x = N\varepsilon \geq 1.$$

Recall that $P_N(N-m) = \frac{1}{2}[f_N(m) + f_N(N-m)]$. Some numerical statements also use odd N , and there x_N is the value of x at which the 2 central bins are as likely as the ends (it is unique by Proposition 2.3). For a mode $a^* \geq N/2$ of P_N we put $m^* = N - a^*$, and m_h is the largest mode of f_N .

3.1 Families and the first flip

For $k_0 \geq 1$ let $(M_k^{[k_0]})_{k \geq k_0}$ be the Markov chain with $M_{k_0}^{[k_0]} = 0$ and the transition probabilities $r_k(b)$ of M . It is the count of disagreeing steps for a walk whose first k_0 steps are forced to agree. So $M^{[1]}$ has the law of M . We write $Y_d = M_d^{[1]}$, and $Z_t^{(j)} = M_t^{[j-1]}$ for $j \geq 2$ and $t \geq j-1$. In a random recursive tree on $[N] = \{1, \dots, N\}$, vertex $k+1$ is joined to a uniform vertex of $[k]$, independently for $k = 1, \dots, N-1$. We write $\mathcal{T}_v$ for the subtree of v , that is, v and its descendants, and we call it the family of v .

Lemma 3.1 (tree and marks). *Let $\mathcal{T}$ be the tree on $[N]$ in which the parent of $k+1$ is U_k . For $2 \leq v \leq N$ let $\mu_v = 1$ if step v flips the step it copies, and $\mu_v = 0$ otherwise. Then $\mathcal{T}$ is a random recursive tree, the marks μ_v are i.i.d. Bernoulli(ε) and independent of $\mathcal{T}$, and $X_v = X_1(-1)^{\sigma(v)}$, where $\sigma(v)$ is the number of marks on the path from v to the root, the root excluded. So $M_N = \#\{v : \sigma(v) \text{ odd}\}$. If the marks on $2, \dots, k_0$ are set to 0, then the number of odd vertices in $[k]$, $k \geq k_0$, is a copy of the chain $M^{[k_0]}$.*

Proof. The parent U_k of $k+1$ is uniform on $[k]$ and independent of the earlier choices, and the marks are the flip indicators. The path from $k+1$ to the root is $k+1$ followed by the path from U_k , so $\sigma(k+1) = \mu_{k+1} + \sigma(U_k)$ and, by induction on k , $X_{k+1} = X_{U_k}(-1)^{\mu_{k+1}} = X_1(-1)^{\sigma(k+1)}$. Now set $\mu_2 = \dots = \mu_{k_0} = 0$. If b of the vertices in $[k]$ are odd, $k \geq k_0$, then $k+1$ is odd with conditional probability $(1-\varepsilon)\frac{b}{k} + \varepsilon(1-\frac{b}{k}) = r_k(b)$, whatever else is known about $[k]$. $\square$

So a single flip at v turns every step of its family. Kürsten's coupling [6] uses bond percolation with retention probability $1 - 2\varepsilon$ instead, which gives the same law of the walk. For $k \geq v$, if $\mathcal{T}_v$ contains s of the vertices in $[k]$, then $k+1$ joins $\mathcal{T}_v$ with probability s/k . This is the urn of

Lemma 2.4, so $|\mathcal{T}_v|$ has the law β_v and mean N/v , and on average there are $N/(d(d+1))$ families of size d . If v is the only marked vertex, the odd vertices are those of $\mathcal{T}_v$. Two or more marks have probability $O(\varepsilon^2)$ at fixed N , and no mark gives $M_N = 0$. So at fixed N the part of $f_N(m)$ of first order in ε is exactly $\varepsilon \sum_v \beta_v(m) = x/(m(m+1))$, for $1 \leq m \leq N-1$ (at $m = h$ this is $c_1 \varepsilon$ of Proposition 2.5). This is also proved in Lean (Section 5).

To go beyond first order we cut the tree at the first flip.

Lemma 3.2 (splitting). *Let $\mathcal{T}$ be a random recursive tree on $[N]$, let $2 \leq v \leq N$ and let $S \subseteq [N]$ with $\min S = v$. Given that $\mathcal{T}_v$ has vertex set S , the restrictions of $\mathcal{T}$ to S and to $[N] \setminus S$, relabeled in increasing order, are independent random recursive trees. They are independent of the parent of v , which is uniform on $[v-1]$.*

Proof. The tree $\mathcal{T}$ is uniform over the $(N-1)!$ increasing trees on $[N]$. Its family $\mathcal{T}_v$ has vertex set S exactly when every $u \in S \setminus \{v\}$ has its parent in S and every $u \notin S$ with $u \geq 2$ has its parent outside S . So, given this event, each element of S other than v picks its parent freely among the smaller elements of S , v picks any element of $[v-1]$, and each element of $[N] \setminus S$ other than 1 picks freely among the smaller elements of $[N] \setminus S$. This is the claim. $\square$

Lemma 3.3 (the first flip). *Let $V = \min\{v \geq 2 : \mu_v = 1\}$, with $V = \infty$ if there is no mark, and let $D = |\mathcal{T}_V|$. Then $\mathbb{P}(V = j, D = d) = \varepsilon(1-\varepsilon)^{j-2} \beta_j(d)$, and for $m \geq 1$*

$$f_N(m) = \sum_{j=2}^N \varepsilon(1-\varepsilon)^{j-2} \sum_{d=1}^{N-j+1} \beta_j(d) \mathbb{P}(d - Y_d + Z_{N-d}^{(j)} = m),$$

where Y_d and $Z_{N-d}^{(j)}$ are independent.

Proof. Marks and tree are independent, which gives the law of (V, D) . Condition on $V = j$ and on the vertex set S of $\mathcal{T}_j$, with $|S| = d$. The marks on $j+1, \dots, N$ are still i.i.d., and by Lemma 3.2 the trees on S and on $[N] \setminus S$ are independent random recursive trees. The vertices $1, \dots, j-1$ carry no mark, so j is odd, and $u \in S$ is odd exactly when the path from u up to j , with j excluded, has an even number of marks. So by Lemma 3.1 for the tree on S rooted at j , the number of odd vertices in S has the law of $d - Y_d$. Paths from outside S stay outside S , and the tree on $[N] \setminus S$ has the unmarked vertices $1, \dots, j-1$ first. So by Lemma 3.1 its odd count has the law of $Z_{N-d}^{(j)}$. The 2 counts are independent, and their law depends on S only through d . Sum over j and d , and note that $M_N = 0$ if there is no mark. $\square$

In words, the first flip turns its whole family except the steps of that family that flip back, and the rest of the tree is a walk whose first $j-1$ steps agree with X_1 . Only the law of each pair $(Y_d, Z_{N-d}^{(j)})$ enters the lemma. So from now on $(Y_d)_{d \geq 1}$ is one path of $M^{[1]}$ and $(Z_t^{(j)})_{t \geq j-1}$ one path of $M^{[j-1]}$, independent of each other and of (V, D) .

Lemma 3.4 (unimodality given the first step). *The function f_N is unimodal.*

This is shown in the proof of Theorem 1.1 of Guérin, Laulin, Raschel and Simon [10], for every N and every memory parameter $1 - \varepsilon$, and it is also proved in Lean (Section 5). So f_N is nondecreasing on $\{0, \dots, m_h\}$ and nonincreasing on $\{m_h, \dots, N\}$. With a fair first step, P_N mixes f_N and its mirror image, whose modes differ. This is where the humps come from.

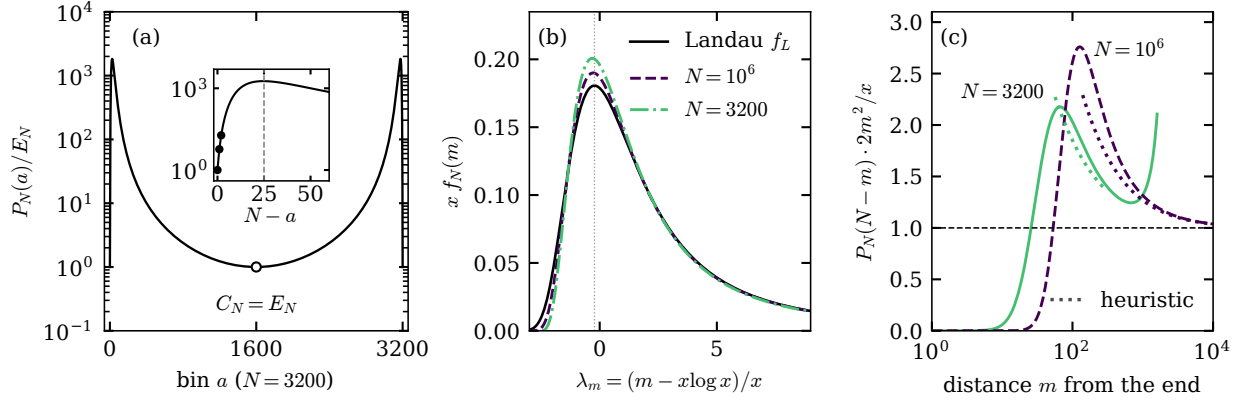


Figure 2: The elephant walk at its crossing. (a) The law P_N at $N = 3200$ and its crossing $x = 11.5468$, divided by the end E_N . The center equals the end, and the law rises from each end to a mode and falls to the center. The inset shows the last 60 bins, with dots at bins N , $N - 1$ and $N - 2$. The end is below its neighbor, and the modes are 25 steps from each end (dashed). (b) The law of M_N , scaled as $x f_N(m)$, against $\lambda_m = (m - x \log x)/x$, with the Landau density f_L . The dotted line is its mode λ_0 . (c) The tail $P_N(N - m)$ divided by $x/(2m^2)$. The dashed line is the limit 1 of Theorem 3.8, and the dotted curves are the heuristic correction $1 + x(2 \log m + 2\gamma - 3)/m$ for $2x \log x \leq m \leq N/x$. The rise of the $N = 3200$ curve near $m = N/2$ is the mirror term $f_N(N - m)$. At $N = 10^6$ we use $x = g_N = 22.4407$.

3.2 The end and its neighbor

Here is why the end should fall below its neighbor once $x > 2$. Given $X_1 = +1$, the end needs no flip at all. Its neighbor, bin $N - 1$, needs 1 flip that is never copied. A flip at step j is never copied with probability about j/N . Summing $\varepsilon j/N$ over j gives a factor of about $x/2$ in favor of the neighbor.

Theorem 3.5 (endpoint dip). *Let $N \geq 3$ and $0 < \varepsilon \leq \frac{1}{2}$, and put $\rho_1 = P_N(N - 1)/E_N$. Then*

$$\frac{x}{2(1 - \varepsilon)} < \rho_1 \leq \frac{x(1 + 1/N)^2}{2 - 3\varepsilon} + \frac{\varepsilon}{1 - \varepsilon}.$$

Hence the end is strictly below its neighbor if $x \geq 2(1 - \varepsilon)$, and strictly above it if the right side is below 1.

Proof. By Lemma 2.2, $\rho_1 = (f_N(1) + f_N(N - 1))/f_N(0)$ with $f_N(0) = (1 - \varepsilon)^{N-1}$. The term $f_N(1)$. On the event $M_N = 1$ the chain moves up once, from time $j - 1$ to time j for some $2 \leq j \leq N$, and then stays at 1. Since $r_t(0) = \varepsilon$ and $1 - r_k(1) = (1 - \varepsilon)(1 - c/k)$ with $c = (1 - 2\varepsilon)/(1 - \varepsilon) \in [0, 1]$,

$$f_N(1) = \varepsilon(1 - \varepsilon)^{N-2} \sum_{j=2}^N P_j, \quad P_j = \prod_{k=j}^{N-1} \left(1 - \frac{c}{k}\right).$$

Since $c \leq 1$, $P_j \geq \prod_{k=j}^{N-1} (1 - 1/k) = (j - 1)/(N - 1)$, and these lower bounds sum to $N/2$. So $f_N(1)/f_N(0) \geq x/(2(1 - \varepsilon))$. Next, $1 - c/k \leq e^{-c/k}$ and $\sum_{k=j}^{N-1} 1/k \geq \log(N/j)$ give $P_j \leq (j/N)^c$. As $(t/N)^c$ is nondecreasing in t ,

$$\sum_{j=2}^N P_j \leq \int_2^{N+1} \left(\frac{t}{N}\right)^c dt \leq \frac{N(1 + 1/N)^{c+1}}{c + 1} \leq \frac{N(1 + 1/N)^2(1 - \varepsilon)}{2 - 3\varepsilon},$$

using $c \leq 1$ and $1/(c+1) = (1-\varepsilon)/(2-3\varepsilon)$. So $f_N(1)/f_N(0) \leq x(1+1/N)^2/(2-3\varepsilon)$. The term $f_N(N-1)$. The event $M_N = N-1$ needs an up move at every time. The first has probability $r_1(0) = \varepsilon$, and the one at time $k \geq 2$ has probability $r_k(k-1) = 1-\varepsilon - (1-2\varepsilon)/k \in [\frac{1}{2}, 1-\varepsilon]$. So $0 < f_N(N-1) \leq \varepsilon(1-\varepsilon)^{N-2}$. This makes the lower bound strict and gives the term $\varepsilon/(1-\varepsilon)$. $\square$

Corollary 3.6 (the dip at the crossing). *Let $N \geq 4$ be even and suppose that $x_N \geq 2(1-\varepsilon_N)$. Then at the crossing $E_N = C_N < P_N(N-1)$, and P_N is not unimodal.*

Proof. By Proposition 2.3, $\varepsilon_N < \frac{1}{2}$, so Theorem 3.5 gives $P_N(N-1) > E_N = C_N$. Suppose that P_N is nondecreasing on $\{0, \dots, a_0\}$ and nonincreasing on $\{a_0, \dots, N\}$. It is symmetric about h , so we may replace a_0 by $N - a_0$ and assume $a_0 \leq h$. Then $P_N(a) \leq C_N$ for every $a \geq h$, including $a = N-1$, which is a contradiction. $\square$

The hypothesis $x_N \geq 2(1-\varepsilon_N)$ holds for every even $N \geq 1688$. To see this, take $\varepsilon = 2/N$. Then $\varepsilon(1 + \log N) \leq \frac{1}{100}$, so Theorem 2.6(b) gives $C_N \leq \frac{8}{N(N+2)} N^{4/N} e^{10400/N}$. The right side decreases in N and is below $1.4 \cdot 10^{-3}$ at $N = 1688$. Since $\log(1-y) \geq -y - y^2$ for $0 \leq y \leq \frac{1}{2}$, we also have $E_N = \frac{1}{2}(1 - \frac{2}{N})^{N-1} \geq \frac{1}{2}e^{-2-2/N} > 0.06$. So $C_N < E_N$ at $\varepsilon = 2/N$, and Proposition 2.3 gives $\varepsilon_N > 2/N$. Hence $x_N > 2 \geq 2(1-\varepsilon_N)$.

Below $N = 1688$ we compute. The hypothesis says $\varepsilon_N \geq 2/(N+2)$, so by Proposition 2.3 it holds exactly when $C_N \leq E_N$ at $\varepsilon = 2/(N+2)$. We evaluated C_N/E_N there for every $8 \leq N < 1688$ in double precision, by 2 independent recursions for the law of M_N (`verify_elephant_shape.py`). They agree, and the largest value is 0.947, far from 1. So Corollary 3.6 holds for every even $N \geq 8$, computer-assisted below $N = 1688$. By Lemma 3.4, P_N is a mixture of 2 unimodal laws with different modes, as is the limit law in [10, eq. (1)].

3.3 The bulk

The steps that disagree with X_1 lie in the flipped families. Let CP_N be the law of $\sum_{d=1}^{N-1} d\mathcal{N}_d$, where the $\mathcal{N}_d$ are independent Poisson variables with means $x/(d(d+1))$. On average $x/(d(d+1))$ flipped vertices have a family of size d . If no 2 flips fall in the same family, which is likely when $x^2 \log N/N$ is small, then M_N is the total size of the flipped families, and this is close to the compound Poisson sum. The families of size up to about x add up to nearly $x \log x$, and the few of size of order x give fluctuations of order x .

Remark 3.7 (compound Poisson and Landau limits, proofs omitted). For $N \geq 2$ and $0 < \varepsilon \leq \frac{1}{4}$, Fourier inversion, the tree of Lemma 3.1 and the Efron–Stein inequality [20] give $\sup_m |f_N(m) - \text{CP}_N(m)| \leq \delta_B := \frac{x^2 \log N}{N} + \frac{4x^2}{N} e^{4x^2/N} + \frac{4x^2(H_N^2 + H_N)}{N}$. Let f_L be the Landau density, with $\int e^{-u\lambda} f_L(\lambda) d\lambda = u^u$ for $u > 0$, and put $\lambda_m = (m - x \log x)/x$. Under the standing assumptions of this section, comparing characteristic functions gives $\sup_m |x f_N(m) - f_L(\lambda_m)| \leq C(\frac{1+\log x}{x} + \frac{x}{N}) + x\delta_B$ with an absolute constant C . So $(M_N - x \log x)/x$ tends to the Landau law when $x \rightarrow \infty$ and $x^3 \log^2 N/N \rightarrow 0$, in particular at the crossing. At $N = 10^6$ and $x = 22.4407$ the bound δ_B is still about 0.46, but there $|f_N(m)/\text{CP}_N(m) - 1| \leq 3.5 \cdot 10^{-4}$ for $1 \leq m \leq 10^4$. At every tested crossing m^* equals the mode of CP_N . If $f_L''(\lambda_0) < 0$ at the mode $\lambda_0 = -0.2228$ of f_L , then the unimodality of f_L [19] and the bound above give $m^* = x \log x + \lambda_0 x + O(\sqrt{x \log x})$ at the crossing for large N . We computed $f_L''(\lambda_0) = -0.0789$ by quadrature but did not certify its sign.

3.4 The tail

Theorem 3.8 (the tail). *Let N be even, and let $x = N\varepsilon \geq 1$ depend on N with $x \log^2 N/N \rightarrow 0$. Let $\omega_N \rightarrow \infty$ with $\omega_N x \log^2(x+1) = o(N)$, and let $\delta'_N \rightarrow 0$. Then*

$$f_N(m) = \frac{x}{m(m+1)}(1 + o(1)) \quad \text{and} \quad P_N(N-m) = \frac{x}{2m^2}(1 + o(1)),$$

uniformly for $\omega_N x \log^2(x+1) \leq m \leq \delta'_N N$.

In words, the chance of landing at distance m from an end is the chance that a flipped family has m steps, halved because the first step must point to that end.

Idea of the proof. The proof is in Appendix B. For the lower bound we use the first flipped family, which has size m with probability about $x/(m(m+1))$ (Lemma B.4), while the other flips change the count only by a relative amount of order $x \log m/m$. For the upper bound we show $\mathbb{P}(M_N \geq m) \leq \frac{x}{m}(1 + O(x \log^2 m/m))$, and the unimodality of f_N turns this into a local bound. The mirror term $f_N(N-m)$ is at most $C_N = O(x/N^2)$, which gives the constant $\frac{1}{2}$.

At $N = 10^6$, $x = 22.4407$ and $m = 10^4$ the exact ratio $P_N(N-m)/(x/m^2)$ is 0.51889. At moderate m , CP_N suggests $f_N(m)m(m+1)/x \approx 1 + x(2 \log m + 2\gamma - 3)/m$, and this factor is still about 2 near $m = 70$ when $N = 1600$ (Figure 2(c)).

3.5 The center

Theorem 3.9 (shape at the crossing). *There are absolute constants C_w and N_0 with the following property. Let $N \geq N_0$ be even with $x_N \in [2, N^{1/4}]$. Then at the crossing $P_N(a) > C_N$ for every $1 \leq a \leq N-1$ with $|a-h| \geq C_w(Nx_N \log N)^{1/2}$.*

By Theorem 2.7, $x_N \in [2, N^{1/4}]$ for large N , so at the crossing the window has width of order $N^{1/2} \log N$. In words, the middle bin is lower than every interior bin except possibly those within about $N^{1/2} \log N$ of it.

Idea of the proof. Up to the mode of f_N the law is above $f_N(1)/2$, which beats $C_N = E_N$ by Theorem 3.5. Farther in, it is close to $x/(2m^2)$, far above the center $4x/N^2$. Near the middle, the main term $x\Pi(m)$ below exceeds its value at h by a factor of about $1 + 12t^2/N^2$ at distance t from h , while we know C_N and the law only up to relative errors of order $x \log N/N$. The 2 meet at t of order $(Nx \log N)^{1/2}$.

Proof. Write $x = x_N$ and $\varepsilon = \varepsilon_N$, so that $C_N = E_N$. By symmetry we take $a = N-m$ with $1 \leq m \leq h$, and put $t = h-m$. Since $2 \leq x \leq N^{1/4}$, every error term below is bounded by a function of N alone, and we take N large. Put $\Pi(m) = \frac{1}{2}[\frac{1}{m(m+1)} + \frac{1}{(N-m)(N-m+1)}]$, so that $x\Pi(h) = \frac{4\varepsilon}{N+2}$. As $\varepsilon(1 + \log N) \rightarrow 0$, Theorem 2.6(b) gives $C_N \leq x\Pi(h)(1 + \eta)$ with $\eta = N^{2\varepsilon}e^{5200\varepsilon} - 1$. Put $u = \varepsilon(2 \log N + 5200)$. Then $u \leq \frac{1}{2}$ for large N , and $e^u - 1 \leq 2u$ for $0 \leq u \leq \frac{1}{2}$, so $\eta \leq 2u \leq 10^4 \varepsilon \log N$.

Step 1: $m \leq m_h$. Then $f_N(m) \geq f_N(1)$ by Lemma 3.4. The proof of Theorem 3.5 gives $f_N(1) \geq \frac{x}{2(1-\varepsilon)}f_N(0)$, and $f_N(0) = 2E_N = 2C_N$. So $P_N(N-m) \geq \frac{1}{2}f_N(m) \geq \frac{x}{2(1-\varepsilon)}C_N > C_N$, since $x \geq 2$.

Step 2: $m_h < m \leq N/8$. Then $f_N(m) \geq f_N(k)$ with $k = \lfloor N/8 \rfloor$, by Lemma 3.4. Proposition B.5(a) at k gives $f_N(k) \geq \frac{x}{k(k+1)}(1 - O(\frac{x \log N}{N}))$. So $P_N(N-m) \geq \frac{32x}{N(N+8)}(1 - o(1))$, which exceeds $\frac{4x}{N(N+2)}(1 + \eta) \geq C_N$.

Step 3: $N/8 < m \leq h$. Then m and $N - m$ are at least $N/8$, and Proposition B.5 at m and at $N - m$ gives $P_N(N - m) \geq x\Pi(m)(1 - \bar{e})$ with $\bar{e} = O(x \log N/N)$. Put $\bar{N} = N + 1$ and $s = 2t/\bar{N}$, so that $m + \frac{1}{2} = \frac{\bar{N}}{2}(1 - s)$ and $N - m + \frac{1}{2} = \frac{\bar{N}}{2}(1 + s)$. Since $\frac{1}{k(k+1)} \geq (k + \frac{1}{2})^{-2}$ and $\Pi(h) = \frac{1}{h(h+1)} = (\frac{\bar{N}}{2})^{-2}(1 - \bar{N}^{-2})^{-1}$, we get

$$\frac{\Pi(m)}{\Pi(h)} \geq \frac{1}{2}[(1 - s)^{-2} + (1 + s)^{-2}](1 - \bar{N}^{-2}) \geq (1 + 3s^2)(1 - \bar{N}^{-2}),$$

because $\frac{1}{2}[(1 - s)^{-2} + (1 + s)^{-2}] = \sum_{k \geq 0} (2k + 1)s^{2k}$. With $b = \bar{e} + \bar{N}^{-2} \leq \frac{1}{2}$ this gives $P_N(N - m) \geq x\Pi(h)(1 + 3s^2)(1 - b) \geq x\Pi(h)(1 + \frac{3}{2}s^2 - b)$.

Step 4: the window. The bound of Step 3 exceeds $x\Pi(h)(1 + \eta) \geq C_N$ as soon as $\frac{3}{2}s^2 > \eta + b$. Now $\eta + b \leq A_2 x \log N/N$ for an absolute constant A_2 , since $x \geq 2$, and $s \geq t/N$. So the condition holds when $t \geq C_w(Nx \log N)^{1/2}$ with $C_w^2 > 2A_2/3$. $\square$

We did not compute C_w or N_0 . Inside the window the proof gives nothing, and closing it needs second differences of the law.

Conjecture 3.10 (convexity between the humps). *For every $N \geq 10$ and $x = x_N$, the second difference $P_N(a - 1) - 2P_N(a) + P_N(a + 1)$ is positive for $2m^* + 1 \leq a \leq N - 2m^* - 1$.*

With the symmetry of the law, the conjecture makes P_N strictly convex on $[2m^*, N - 2m^*]$, so h is its unique minimizer there (the 2 central bins for odd N). The next corollary shows that $m^* = o(N)$ at the crossing, so together with Theorem 3.9 the conjecture would make the center the unique interior minimum for large even N .

Corollary 3.11 (the modes). *For large even N the modes a^* of P_N at the crossing satisfy $\min(a^*, N - a^*) = O(x_N \log^2 x_N)$.*

Proof. Write $x = x_N$ and $\varepsilon = \varepsilon_N$. By Theorem 2.7, $x \sim 2 \log N$, so for large N the hypotheses of Theorem 2.6(b) and Lemma B.7 hold. The law is symmetric about h , so $m^* = \min(a^*, N - a^*)$ maximizes $f_N(m) + f_N(N - m)$ over $0 \leq m \leq h$. Lemma B.7 gives $m_h \leq h$ for large N . So Lemma 3.4 and Theorem 2.6(b) give $f_N(N - m) \leq f_N(h) = C_N = O(x/N^2)$ for $m \leq h$, and hence $f_N(m^*) \geq f_N(m_h) - C_N$. Let ℓ be as in the proof of Lemma B.7. Then $f_N(m_h) \geq f_N(\ell) \geq \frac{x}{2\ell(\ell+1)}$, so $f_N(m^*) \geq \frac{x}{4\ell(\ell+1)}$ for large N . If $m^* > m' = \max(m_h, \ell)$, then Lemmas B.1 and B.2 give $(m^* - m' + 1)f_N(m^*) \leq \mathbb{P}(M_N \geq \ell) \leq xH_\ell/\ell$. Hence $m^* \leq \max(m_h, \ell) + 4(\ell + 1)H_\ell$ in every case. Finally Lemma B.7 gives $m_h = O(x \log^2 x)$, and $\ell = O(x \log x)$ and $H_\ell = O(\log x)$. $\square$

The conjecture holds for every $10 \leq N \leq 600$ and for $N = 800, 1600, \dots, 25600$, and on $[2m^*, N - 2m^*]$ it fails only at the 2 endpoints for some $N \leq 51$. Heuristically, a concave stretch of f_N far from its mode is added to the tail $x/(2r^2)$ of the other branch, with $r = N - a$, and the second difference $3x/r^4$ of this tail beats the concave curvature, which is of order $x^2/(N^2 r^3)$. We leave open the window around the center, the tail between $x \log x$ and $x \log^2 x$, and effective constants in Theorems 3.8 and 3.9.

4 Heavy-tailed runs

In this section the walk remembers the time since its last switch. The steps form maximal runs whose directions alternate. The first run has length D , and the later runs have i.i.d. lengths $\xi_1, \xi_2, \dots$ distributed as ξ , where

$$\bar{F}(k) = \mathbb{P}(\xi \geq k) = k^{-\alpha} \quad (k \geq 1)$$

for some $\alpha > 0$. So $\mathbb{P}(\xi = k) = k^{-\alpha} - (k+1)^{-\alpha} > 0$ for every $k \geq 1$. The first direction is fair, and D , (ξ_i) and the first direction are independent. With a fresh start D has the law of ξ . For $\alpha > 1$ the stationary start is $\mathbb{P}(D = d) = \bar{F}(d)/\mu$, where $\mu = \mathbb{E}\xi = \zeta(\alpha)$. This law is stationary in the following sense. The number of steps left in the current run, counting the current one, is a Markov chain that moves from $k \geq 2$ to $k-1$ and from 1 to a fresh copy of ξ . Since $\bar{F}(k) = \bar{F}(k+1) + \mathbb{P}(\xi = k)\bar{F}(1)$, the law $\bar{F}(k)/\mu$ is invariant for this chain. So the stationary start sees the runs at a typical time. We define A_N , P_N , C_N and E_N as in Section 1.2, put $R_N = E_N/C_N$, and write $\mathbb{P}_+$ for the law given that the first run is $+$. In this section the constants implied by $O(\cdot)$ may depend on α .

Proposition 4.1 (the edge). *For every law of D , $E_N = \frac{1}{2}\mathbb{P}(D \geq N)$. With a fresh start $E_N = \frac{1}{2}N^{-\alpha}$. With the stationary start*

$$E_N = \frac{\zeta(\alpha, N)}{2\zeta(\alpha)} = \frac{N^{1-\alpha}}{2(\alpha-1)\zeta(\alpha)} \left(1 + \frac{\theta_N(\alpha-1)}{N}\right)$$

for some $\theta_N \in [0, 1]$, where $\zeta(\alpha, N) = \sum_{k \geq N} k^{-\alpha}$ is the Hurwitz zeta function.

Proof. All N steps are $+1$ exactly when the first direction is $+$ and $D \geq N$. Hence $E_N = \frac{1}{2}\mathbb{P}(D \geq N)$, and $\mathbb{P}(\xi \geq N) = N^{-\alpha}$. For the stationary start $\mathbb{P}(D \geq N) = \zeta(\alpha, N)/\zeta(\alpha)$. Since $t^{-\alpha}$ is decreasing, $\int_N^\infty t^{-\alpha} dt \leq \zeta(\alpha, N) \leq N^{-\alpha} + \int_N^\infty t^{-\alpha} dt$, and the integral equals $N^{1-\alpha}/(\alpha-1)$. $\square$

So the edge only sees the first run, while the center depends on all the runs. For the center we need the renewal quantities

$$\begin{aligned} \tau_m &= \xi_1 + \dots + \xi_m, \quad \tau_0 = 0, \quad u_m(k) = \mathbb{P}(\tau_m = k), \\ q_m(k) &= \mathbb{P}(\tau_{m-1} < k \leq \tau_m) = \sum_{j \geq 1} \bar{F}(j) u_{m-1}(k-j) \end{aligned}$$

for $m, k \geq 1$. So $u_m(k)$ is the chance that the m -th run ends at level k , and $q_m(k)$ the chance that the m -th run covers level k . For the delayed versions let D be independent of (ξ_i) , and put $u_m^D(k) = \mathbb{P}(D + \tau_{m-1} = k)$, $q_1^D(k) = \mathbb{P}(D \geq k)$ and $q_m^D(k) = \mathbb{P}(D + \tau_{m-2} < k \leq D + \tau_{m-1})$ for $m \geq 2$. For the stationary start $u_m^D(k) = \sum_d \frac{\bar{F}(d)}{\mu} u_{m-1}(k-d) = q_m(k)/\mu$.

The $+$ runs and the $-$ runs form 2 independent renewal processes. The walk is in bin a at time N exactly when one of them sits at its level and the other is inside a run that covers its level.

Proposition 4.2 (two renewal processes). *Let D have any law on $\{1, 2, \dots\}$. For $a + b = N$ with $a, b \geq 1$,*

$$\mathbb{P}_+(A_N = a) = \sum_{m \geq 1} \left[u_m(b) q_{m+1}^D(a) + u_m^D(a) q_m(b) \right],$$

and $\mathbb{P}_+(A_N = N) = \mathbb{P}(D \geq N)$. In particular, for even N , $C_N = \sum_{m \geq 1} [u_m(h) q_{m+1}^D(h) + u_m^D(h) q_m(h)]$. For the fresh start this is $C_N = \sum_m u_m(h) [q_m(h) + q_{m+1}(h)]$.

Proof. Let the $+$ runs have lengths $D, \xi_2^+, \xi_3^+, \dots$ and the $-$ runs $\xi_1^-, \xi_2^-, \dots$, and let $s_j^\pm$ be the sum of the first j lengths of each sign, with $s_0^\pm = 0$. So s_j^+ has law u_j^D and s_j^- has law u_j , and the 2 sequences are independent. The j -th $+$ run covers the steps in $(s_{j-1}^+ + s_{j-1}^-, s_j^+ + s_{j-1}^-]$, and the j -th $-$ run covers $(s_j^+ + s_{j-1}^-, s_j^+ + s_j^-]$. We split according to the run that contains step $N = a + b$. If it is the first run, then $A_N = N$, and this happens exactly when $D \geq N$. If it is the

j -th + run with $j \geq 2$, then $A_N = N - s_{j-1}^-$, so $A_N = a$ means $s_{j-1}^- = b$ and $s_{j-1}^+ < a \leq s_j^+$. This has probability $u_{j-1}(b)q_j^D(a)$. If it is the j -th - run, then $A_N = s_j^+$, so $A_N = a$ means $s_j^+ = a$ and $s_{j-1}^- < b \leq s_j^-$. This has probability $u_j^D(a)q_j(b)$. The cases are disjoint. Summing them, with $m = j - 1$ in the second and $m = j$ in the third, gives the formula. For even N the symmetry gives $C_N = \frac{1}{2}[\mathbb{P}_+(A_N = h) + \mathbb{P}_+(A_N = N - h)] = \mathbb{P}_+(A_N = h)$. $\square$

This identity is the coefficient form of the double transform of Godrèche and Luck [11, Section 7] for the occupation time of an alternating renewal process. To evaluate it we need the stable limit of τ_m and its local form.

Lemma 4.3 (stable and local limits). *Let $1 < \alpha < 2$ and $a_m = m^{1/\alpha}$. Put $K_\alpha = \Gamma(2-\alpha)/(\alpha-1) = -\Gamma(1-\alpha) > 0$ and $\sigma_\alpha = 2K_\alpha|\cos \frac{\pi\alpha}{2}|$.*

- (i) $\mathbb{E}e^{i\theta\xi} = 1 + i\mu\theta + K_\alpha(-i\theta)^\alpha + O(\theta^2)$ as $\theta \rightarrow 0$, with the principal branch.
- (ii) $(\tau_m - m\mu)/a_m$ converges in law to a variable G with $\mathbb{E}e^{-sG} = e^{K_\alpha s^\alpha}$ for $s \geq 0$ and $\mathbb{E}e^{i\theta G} = \exp(K_\alpha(-i\theta)^\alpha)$. The law of G is totally skewed to the right and has a bounded, uniformly continuous density g with $g(z) \rightarrow 0$ as $|z| \rightarrow \infty$.
- (iii) (Gnedenko) The constant $\bar{u} := \sup_{m \geq 1} \sup_k a_m u_m(k)$ is finite, and $a_m u_m(k) - g((k - m\mu)/a_m) \rightarrow 0$ uniformly in k as $m \rightarrow \infty$.
- (iv) If G' is an independent copy of G , then $G - G'$ is symmetric α -stable with $\mathbb{E}e^{i\theta(G-G')} = e^{-\sigma_\alpha|\theta|^\alpha}$, and $I_\alpha := \int_{\mathbb{R}} g(z)^2 dz = \Gamma(1 + 1/\alpha)/(\pi\sigma_\alpha^{1/\alpha})$.

Proof. (i) Summation by parts gives $1 - \mathbb{E}w^\xi = (1 - w) \sum_{k \geq 1} \bar{F}(k)w^{k-1}$ for $|w| \leq 1$. With $w = e^{-v}$ and $\text{Li}_\alpha(w) = \sum_{k \geq 1} k^{-\alpha}w^k$ this is $1 - \mathbb{E}e^{-v\xi} = (e^v - 1)\text{Li}_\alpha(e^{-v})$. For $\text{Re } v \geq 0$ and $0 < |v| < 2\pi$, [21, 25.12.12] gives $\text{Li}_\alpha(e^{-v}) = \Gamma(1-\alpha)v^{\alpha-1} + \zeta(\alpha) + O(|v|)$. So $1 - \mathbb{E}e^{-v\xi} = \mu v + \Gamma(1-\alpha)v^\alpha + O(|v|^2)$, and $v = -i\theta$ gives (i). (ii) By (i), $e^{-i\mu t}\mathbb{E}e^{it\xi} = 1 + K_\alpha(-i\theta)^\alpha/m + O(\theta^2 m^{-2/\alpha})$ for $t = \theta/a_m$. As $2/\alpha > 1$, the characteristic function of $(\tau_m - m\mu)/a_m$ tends to $\exp(K_\alpha(-i\theta)^\alpha)$, whose modulus $\exp(-\frac{1}{2}\sigma_\alpha|\theta|^\alpha)$ is integrable. Lévy's continuity theorem, Fourier inversion and the Riemann-Lebesgue lemma give G and g (the domain of attraction also follows from [17, Theorem XVII.5.2]). For real $s > 0$, the expansion of $1 - \mathbb{E}e^{-v\xi}$ above with $v = s/a_m$ gives $\mathbb{E}e^{-s(\tau_m - m\mu)/a_m} \rightarrow e^{K_\alpha s^\alpha}$ in the same way, and with $2s$ in place of s this gives uniform integrability, so $\mathbb{E}e^{-sG} = e^{K_\alpha s^\alpha}$. (iii) Every $\mathbb{P}(\xi = k)$ is positive, so ξ has maximal span 1, and Gnedenko's local limit theorem [16, Chapter 4] gives the limit. Then $a_m u_m(k) \leq \|g\|_\infty + 1$ for large m and $a_m u_m(k) \leq a_m$ for the others, so $\bar{u} < \infty$. (iv) The function $|\mathbb{E}e^{i\theta G}|^2 = e^{-\sigma_\alpha|\theta|^\alpha}$ is the characteristic function of $G - G'$, and Plancherel's theorem gives $I_\alpha = \frac{1}{2\pi} \int e^{-\sigma_\alpha|\theta|^\alpha} d\theta$, which is $\Gamma(1 + 1/\alpha)/(\pi\sigma_\alpha^{1/\alpha})$. $\square$

Theorem 4.4 (the center for $1 < \alpha < 2$). *Let $1 < \alpha < 2$ and let D be any a.s. finite first run, in particular either start. As $N \rightarrow \infty$ over even N ,*

$$C_N \sim 2I_\alpha \left(\frac{\mu}{h}\right)^{1/\alpha} = \frac{2\Gamma(1 + 1/\alpha)}{\pi} \left(\frac{\mu}{K_\alpha|\cos(\pi\alpha/2)|N}\right)^{1/\alpha}.$$

In words, the center decays like $N^{-1/\alpha}$, and the first run does not matter.

α	$e(\alpha)$	$C(\alpha)$	$C_{\text{fr}}(\alpha)$	R_{1600}	R_{10^4}	R_{10^5}
1.3	+0.469231	0.422717	0.498631	17.7593 [1.3180]	38.7537 [1.2171]	106.149 [1.1317]
1.5	+0.166667	0.647971	0.846371	2.14409 [0.9675]	2.9528 [0.9818]	4.37642 [0.9914]
φ	0	0.776132	1.073675	0.661763 [0.8526]	0.697331 [0.8985]	0.727689 [0.9376]
5/3	-1/15	0.835101	1.182237	0.411815 [0.8064]	0.389307 [0.8614]	0.352822 [0.9102]
1.7	-0.111765	0.880046	1.265508	0.298369 [0.7733]	0.262005 [0.8334]	0.215839 [0.8881]
1.9	-0.373684	1.438986	2.266075	0.0449706 [0.4923]	0.0256505 [0.5569]	0.0121612 [0.6242]

Table 2: Constants of Corollary 4.5 and exact values of $R_N = E_N/C_N$ for the stationary start, with $R_N/(C(\alpha)N^{e(\alpha)})$ in brackets.

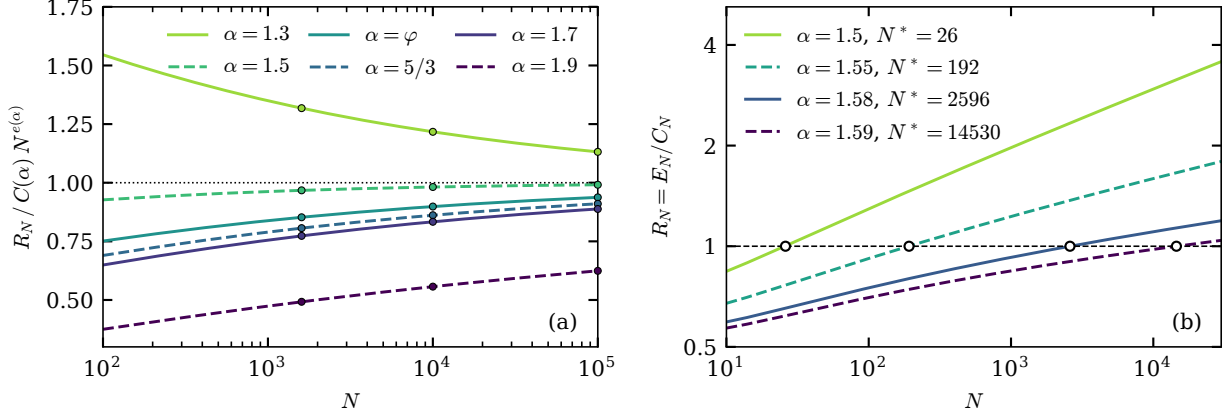


Figure 3: Alternating runs with a stationary start. (a) $R_N/(C(\alpha)N^{e(\alpha)})$ for even N from 100 to 10^5 , with the entries of Table 2 as dots. (b) R_N for $\alpha = 1.5, 1.55, 1.58, 1.59$ below φ . Each curve crosses 1 once on $10 \leq N \leq 3 \cdot 10^4$, at $N^*(\alpha)$ (circles).

Idea of the proof. Both kinds of runs must fill about h steps, which takes about h/μ runs of each kind. Each renewal process is then within about $N^{1/\alpha}$ of its mean, so the 2 meet at a given level with probability of order $N^{-1/\alpha}$. In the proof (Appendix C) we split the sum in Proposition 4.2 by the number m of runs. With $b_h = (h/\mu)^{1/\alpha}$, the terms with $|m\mu - h| \leq Mb_h$ form a Riemann sum for $2b_h^{-1} \int_{-M}^M g^2$ by Lemma 4.3(iii). The terms with $m \leq h/(4\mu)$ add up to $O(h^{-\alpha}) = o(b_h^{-1})$, since a large sum of few runs is most likely made by one long run. The others add up to $O(b_h^{-1})\mathbb{P}(|G| \geq M-1) + o(b_h^{-1})$, and we let $M \rightarrow \infty$.

Corollary 4.5 (the golden-ratio exponent). *Let $1 < \alpha < 2$ and $e(\alpha) = 1 - \alpha + 1/\alpha$. If $\mathbb{P}(D \geq N) \sim c_D N^{-\beta_D}$ with $c_D, \beta_D > 0$, then $R_N \sim c_D N^{1/\alpha - \beta_D} / (4I_\alpha(2\mu)^{1/\alpha})$. In particular $R_N \sim C(\alpha)N^{e(\alpha)}$ for the stationary start and $R_N \sim C_{\text{fr}}(\alpha)N^{e(\alpha)-1}$ for the fresh start, where*

$$C(\alpha) = \frac{\pi(K_\alpha |\cos \frac{\pi\alpha}{2}|)^{1/\alpha}}{4(\alpha-1)\Gamma(1+\frac{1}{\alpha})\zeta(\alpha)^{1+1/\alpha}}, \quad C_{\text{fr}}(\alpha) = (\alpha-1)\zeta(\alpha)C(\alpha).$$

Since $e(\alpha) > 0$ exactly when $\alpha < \varphi$, with a stationary start the ends win for all large N if $\alpha < \varphi$, and the center wins if $\alpha > \varphi$. At $\alpha = \varphi$,

$$R_N \rightarrow C(\varphi) = \frac{\pi\varphi(\varphi\Gamma(\varphi^{-2})|\cos \frac{\pi\varphi}{2}|)^{1/\varphi}}{4\Gamma(\varphi)\zeta(\varphi)^\varphi} = 0.7761317206\dots,$$

α	start	center C_N	ratio R_N	status
$\alpha > 2$	stationary	$\sim \sqrt{2\mu/(\pi v N)}$	$\sim C_2(\alpha)N^{3/2-\alpha}$	same argument, omitted
$\alpha = 2$	stationary	$\sim \sqrt{2\mu/(\pi N \log N)}$	$\sim \sqrt{\pi \log N/(8\zeta(2)^3 N)}$	same argument, omitted
$1 < \alpha < 2$	fresh	$\sim 2I_\alpha(\mu/h)^{1/\alpha}$	$\sim C_{\text{fr}}(\alpha)N^{e(\alpha)-1} \rightarrow 0$	Corollary 4.5
$\alpha = 1$	fresh	$\approx 2/(\pi^2 \tilde{m}_h)$	$\approx \pi^2 \tilde{m}_h/(8h) \sim \pi^2/(8 \log N)$	heuristic
$0 < \alpha < 1$	fresh	$\sim \frac{2}{\pi} \tan \frac{\pi\alpha}{2} \cdot \frac{1}{N}$	$\sim \frac{\pi}{4 \tan(\pi\alpha/2)} N^{1-\alpha} \rightarrow \infty$	proof omitted

Table 3: Outcomes of Remark 4.6. The center wins when $R_N \rightarrow 0$ and the ends win when $R_N \rightarrow \infty$.

so the center still wins by the factor $1/C(\varphi) = 1.2884$. With a fresh start $e(\alpha) - 1 < 0$, and the center wins for every $1 < \alpha < 2$.

Proof. By Proposition 4.1 and Theorem 4.4, with $h = N/2$, we have $R_N = \mathbb{P}(D \geq N)/(2C_N)$, and this is asymptotic to $c_D N^{-\beta_D}/(4I_\alpha(2\mu/N)^{1/\alpha})$, which is the general formula. By Lemma 4.3(iv), $4I_\alpha(2\mu)^{1/\alpha} = 4\Gamma(1 + \frac{1}{\alpha})\mu^{1/\alpha}/(\pi(K_\alpha|\cos \frac{\pi\alpha}{2}|)^{1/\alpha})$. The fresh start has $c_D = 1$ and $\beta_D = \alpha$, which gives $C_{\text{fr}}(\alpha)$. By Proposition 4.1 the stationary start has $c_D = 1/((\alpha-1)\mu)$ and $\beta_D = \alpha-1$, which gives $C(\alpha)$. Next, $e(\alpha) = -(\alpha^2 - \alpha - 1)/\alpha$, and φ is the positive root of $\alpha^2 - \alpha - 1$. At $\alpha = \varphi$ we have $\alpha - 1 = 1/\varphi$, $1 + 1/\alpha = \varphi$ and $2 - \alpha = \varphi^{-2}$. So $K_\varphi = \varphi \Gamma(\varphi^{-2})$, and substituting these into $C(\alpha)$ gives the closed form. $\square$

Corollary 4.5 is the lattice form, with its constant, of the heuristic of [15, Section 2.1]. The golden ratio also appears for the elephant walk, as the log-concavity threshold $1/\varphi$ of [10, Proposition 2.2], but we know of no connection. The approach to $C(\varphi)$ is slow. At $\alpha = \varphi$ we have $R_{10^5}/C(\varphi) = 0.938$, and R_N increases on the computed grid (Table 2 and Figure 3).

Remark 4.6 (other exponents). Table 3 collects the other cases. For $\alpha > 2$ let $v = \text{Var } \xi = 2\zeta(\alpha-1) - \zeta(\alpha) - \zeta(\alpha)^2$ and $C_2(\alpha) = \sqrt{\pi v/(2\mu)}/(2(\alpha-1)\mu)$. The proof of Theorem 4.4 works with the Gaussian local limit theorem and $a_m = \sqrt{vm}$, and for $\alpha = 2$ with the norming $a_m \sim \sqrt{m \log m}$ of [17, Theorem XVII.5.3 and eq. (XVII.5.23)]. In these 2 rows the center formula holds for every a.s. finite first run, so a fresh start also gives $R_N \rightarrow 0$. For $\alpha = 1$ the same proof with a skewed Cauchy limit and the centering $m(H_{m+1} - 1)$ suggests the row of the table, where $\tilde{m}_h$ solves $m(H_{m+1} - 1) = h$. For $0 < \alpha < 1$ only the fresh start makes sense, and $\frac{2}{\pi} \tan \frac{\pi\alpha}{2}$ is the density at $\frac{1}{2}$ of Lamperti's occupation-time law [12, 11]. Here the proof splits the sum in Proposition 4.2 on the scale of $\log m$ and uses the moments of a positive stable law [18, Theorem 2.6.3] and the series for its density [17, Lemma XVII.6.1]. We omit the details for $\alpha \geq 2$ and $0 < \alpha < 1$. So with a fresh start the threshold is $\alpha = 1$.

Remark 4.7 (crossover sizes and heat conduction). For $\alpha < \varphi$ and a stationary start the crossover is late when α is close to φ . For $\alpha = 1.5, 1.55, 1.58, 1.59$ and every even N in $[10, 3 \cdot 10^4]$, $R_N - 1$ changes sign once, at $N^*(\alpha) = 26, 192, 2596, 14530$. Heuristically, $C(\alpha)N^{e(\alpha)} = 1$ gives $\log N^*(\alpha) = 0.1834/(\varphi - \alpha) + O(1)$ as $\alpha \uparrow \varphi$. At the value $\alpha = 5/3$ of Lévy-walk models of heat conduction [14, 13], $R_N \sim C(5/3)N^{-1/15}$ with $C(5/3) = 0.8351$.

Remark 4.8 (aging). Let the walk take a fair first step and then switch direction at step $k \geq 2$ with probability c/k , where $0 < c < 2$. Then A_N/N converges in law to Beta(c, c) [22, Example 1.1 and Theorem 1.4] (their chain is this walk for $c \leq 1$), [23, Theorem 1], and the limit is uniform exactly when $c = 1$ [23, Remark 2]. At finite N the first step matters. Let X_N be the last step. At $c = 1$, induction on N gives $\mathbb{P}_+(A_N = s, X_N = +1) = \frac{s-1}{N(N-1)}$ and $\mathbb{P}_+(A_N = s, X_N = -1) = \frac{N-s}{N(N-1)}$ for $1 \leq s \leq N$, so A_N is uniform on $\{1, \dots, N\}$ (after a shift of one step this is the exercise in [24,

footnote 1 on p. 6]). Compared with no switch, a switch at time t multiplies the probability of a path by $c/(t - c)$, which increases in c , and every path with $A_N < N$ switches. So as in Proposition 2.3, $\mathbb{P}_+(A_N = s)/\mathbb{P}_+(A_N = N)$ increases strictly in c for $1 \leq s \leq N - 1$, and the fixed-start crossing of every bin is exactly $c = 1$. With a fair first step, at $c = 1$ the ends are $\frac{1}{2N}$ and every interior bin is $\frac{1}{N}$, so C_N/E_N , which increases strictly in c from 0, has a unique crossing $c_N < 1$. For even N , $c_N = 1 - \log 2/(\log N + \gamma + 2 - 2 \log 2) + O((\log N)^{-3})$, and the exact crossing at $N = 200$ is $c_{200} = 0.8925$, against 0.8932 from the first 2 terms. This follows from $E_N = \frac{1}{2} \prod_{k=2}^N (1 - c/k)$ and a local limit theorem with a rate, $NC_N \rightarrow 4^{1-c}/B(c, c)$ uniformly for c near 1 (details omitted). The exact statements here are proved in Lean too.

We do not know a rate in Theorem 4.4, whether R_N is monotone with a single sign change of $R_N - 1$, how to make the case $\alpha = 1$ rigorous, or the full shape of P_N for heavy-tailed runs.

5 Verification

All computations are in Python, and the main ones were done by 2 independent methods (for example, forward and backward recursions for the elephant center agree to about 10^{-14} at $N = 12800$). The proofs use computation in only 3 places. Some constants are values of explicit sums and integrals, such as the 60-term sum in the proof of Theorem 2.6(b), which we evaluated in 30-digit arithmetic (`verify_elephant_crossing.py`). Remark 3.7 uses the sign of $f_L''(\lambda_0)$, computed by quadrature but not certified. And Corollary 3.6 holds for every even $N \geq 8$, but for $N < 1688$ its hypothesis rests on a double-precision computation of C_N/E_N at $\varepsilon = 2/(N + 2)$, described after the corollary (`verify_elephant_shape.py`). Conjecture 3.10, the tables, the figures and the numbers quoted outside the proofs rest on computation alone. Before each new computation we wrote down a prediction and a condition that would refute it. Of these predictions 8 failed, and none of them concerns a statement that we call a theorem. The scripts, their output and the prediction ledgers are in the Problem 131 folder of <https://github.com/apemm/Kagey-Problems>, where `SUPPLEMENT.md` says what each script checks and `python -B verify_all.py --paper d` reruns them in a few minutes. The exact and finite statements, among them Proposition 2.3, Lemma 3.4, Theorem 3.5 and the identities behind Corollary 4.5, are also proved in Lean 4 in `lean/PaperD`, and `MAP.md` there lists each one with its Lean name.

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A Proof of Theorem 2.6

Throughout, $N \geq 2$ is even and we condition on $X_1 = +1$. By Proposition 2.5(a) we need to bound $V_{t+1}(1)$, and we compare V with explicit hitting probabilities of Pólya urns. Let $\mathcal{L}_s F(b) = (1 - r_s(b))F(b) + r_s(b)F(b + 1)$, so that $V_s = \mathcal{L}_s V_{s+1}$ and $V_N(b) = \mathbf{1}\{b = h\}$. Both M_t and $t - M_t \geq 1$ are nondecreasing. So from $b \geq 1$ the chain stays at states ≥ 1 , and for $b \geq 1$ we have $V_s(b) = 0$ unless b is feasible, that is, $b \in \mathcal{S}_s = \{b \in \mathbb{Z} : 1 \leq b \leq h, 1 \leq s - b \leq h\}$. Note that $\mathcal{S}_N = \{h\}$.

For $c \geq 0$ the Pólya(c) chain moves at time s from b to $b + 1$ with probability $r_s^c(b) = \frac{b+c}{s+2c}$. It is Pólya's urn with weights $b + c$ and $s - b + c$ for the 2 colors, that is, with c extra balls of each color. For $b + c > 0$ and $s - b + c > 0$ let $\Phi_s^c(b) = \binom{N-s}{h-b} B(h + c, h + c) / B(b + c, s - b + c)$, where B

is the beta function and the binomial coefficient is 0 unless $0 \leq h - b \leq N - s$. Theorem 2.6(a) will follow from Lemma A.1(d), and (b) from Proposition A.4.

Lemma A.1. *Let $c \geq 0$, $1 \leq s \leq N$, $b + c > 0$ and $s - b + c > 0$.*

- (a) $\Phi_s^c(b)$ is the probability that the Pólya(c) chain at b at time s is at h at time N . So $\Phi_N^c(b) = \mathbf{1}\{b = h\}$, and $\Phi_s^c(b) = r_s^c(b)\Phi_{s+1}^c(b+1) + (1 - r_s^c(b))\Phi_{s+1}^c(b)$ for $s < N$.
- (b) If $s < N$ and $\Phi_s^c(b) > 0$, then $r_s^c(b)\Phi_{s+1}^c(b+1)/\Phi_s^c(b) = \bar{r}_s(b) := \frac{h-b}{N-s}$.
- (c) If $0 < \varepsilon < \frac{1}{2}$ and $\beta = \frac{\varepsilon}{1-2\varepsilon}$, then $r_s(b) = r_s^{\beta s}(b)$ for $0 \leq b \leq s$.
- (d) If $0 < \varepsilon < \frac{1}{2}$ and $0 \leq c \leq \beta s$, then $V_s(b) \geq \Phi_s^c(b)$ for $1 \leq b \leq s-1$. In particular $V_{t+1}(1) \geq \Phi_{t+1}^0(1) = \pi_t$ for $1 \leq t \leq h$.

Proof. (a) Let $x = b + c$, $y = s - b + c$ and $(x)_k = x(x+1) \cdots (x+k-1)$. As in the proof of Lemma 2.4, each order of the $N - s$ draws with $h - b$ draws of the first color has probability $(x)_{h-b}(y)_{N-s-h+b}/(x+y)_{N-s} = B(h+c, h+c)/B(x, y)$, and there are $\binom{N-s}{h-b}$ such orders. The recursion is the Markov property.

(b) Use $\binom{N-s-1}{h-b-1}/\binom{N-s}{h-b} = \frac{h-b}{N-s}$ and $B(x, y)/B(x+1, y) = \frac{x+y}{x} = 1/r_s^c(b)$.

(c) Since $1 + 2\beta = 1/(1 - 2\varepsilon)$, we have $\frac{b+\beta s}{s+2\beta s} = (1 - 2\varepsilon)(\frac{b}{s} + \beta) = r_s(b)$.

(d) Let $F_u = \Phi_u^c$ on $\{1, \dots, u-1\}$ for $s \leq u \leq N$. Then F_u vanishes off $\mathcal{S}_u$ and $F_N(b) = \mathbf{1}\{b = h\}$. By (a) and (c), $\mathcal{L}_u F_{u+1}(b) - F_u(b) = (r_u(b) - r_u^c(b))(F_{u+1}(b+1) - F_{u+1}(b))$. Since $r_u^c(b) = \frac{1}{2} - (\frac{u}{2} - b)/(u+2c)$, this is the product of $\frac{u}{2} - b$, $\frac{1}{u+2c} - \frac{1}{u(1+2\beta)} \geq 0$ (as $c \leq \beta s \leq \beta u$) and $F_{u+1}(b+1) - F_{u+1}(b)$. Let $b \in \mathcal{S}_u$. If $b = h$ then $b \geq u/2$ and $F_{u+1}(b+1) = 0$. If $u - b = h$ then $b \leq u/2$ and $F_{u+1}(b) = 0$. Otherwise $F_{u+1}(b+1)/F_{u+1}(b) = \frac{(h-b)(u-b+c)}{(h-u+b)(b+c)}$, which is ≥ 1 exactly when $b \leq u/2$ (write $b = \frac{u}{2} - d$; the difference of the 2 sides of the inequality is $2d(h+c)$). So in all 3 cases the first and last factors have the same sign, and $\mathcal{L}_u F_{u+1} \geq F_u$ on $\mathcal{S}_u$, and Lemma A.2(b) gives $V_s \geq F_s$. Finally $\Phi_{t+1}^0(1) = \beta_{t+1}(h) = \pi_t$ by (a) and Lemma 2.4. $\square$

Lemma A.2. *Let $1 \leq s_0 \leq N$, and for $s_0 \leq s \leq N$ let $F_s \geq 0$ be a function on $\{1, \dots, s-1\}$.*

- (a) If $F_N(h) \geq 1$ and $\mathcal{L}_s F_{s+1} \leq K_s F_s$ on $\mathcal{S}_s$ for $s_0 \leq s < N$, with constants $K_s \geq 0$, then $V_s \leq (\prod_{u=s}^{N-1} K_u) F_s$ on $\mathcal{S}_s$.
- (b) If $F_N(b) \leq \mathbf{1}\{b = h\}$, each F_s vanishes off $\mathcal{S}_s$, and $\mathcal{L}_s F_{s+1} \geq F_s$ on $\mathcal{S}_s$ for $s_0 \leq s < N$, then $V_s \geq F_s$.

Proof. We use backward induction on s , from $V_N(b) = \mathbf{1}\{b = h\}$. In (a), V_{s+1} vanishes at infeasible states and $F_{s+1} \geq 0$, so on $\mathcal{S}_s$ we get $V_s = \mathcal{L}_s V_{s+1} \leq \prod_{u>s} K_u \mathcal{L}_s F_{s+1} \leq \prod_{u \geq s} K_u F_s$. In (b) we get $V_s \geq \mathcal{L}_s F_{s+1} \geq F_s$ on $\mathcal{S}_s$, and $F_s = 0 \leq V_s$ off $\mathcal{S}_s$. $\square$

For the upper bound we compare M with an urn that has more extra balls, and we control the cost of each step through the derivative of Φ_s^τ in τ . Let ψ be the digamma function.

Lemma A.3. *Let $2 \leq s \leq N$, $b \in \mathcal{S}_s$ and $\tau \geq 0$. Put $X = b + \tau$, $Y = s - b + \tau$, $\Sigma = X + Y$ and $z = (s - 2b)^2/\Sigma^2$. Then*

$$\partial_\tau \log \Phi_s^\tau(b) < -\log(1 - z) + \frac{1 + z}{(1 - z)\Sigma} + \frac{2(1 + z)}{3\Sigma^2(1 - z)^2}.$$

Proof. Since $\log B(x, y) = \log \Gamma(x) + \log \Gamma(y) - \log \Gamma(x+y)$, the derivative is $2\psi(\Sigma) - \psi(X) - \psi(Y) - 2\psi(N+2\tau) + 2\psi(h+\tau)$. For $x > 0$, [21, 5.9.15] writes $\log x - \frac{1}{2x} - \psi(x)$ as $2 \int_0^\infty \frac{t dt}{(t^2+x^2)(e^{2\pi t}-1)}$, which lies in $(0, \frac{1}{12x^2})$ since $\int_0^\infty \frac{t dt}{e^{2\pi t}-1} = \frac{1}{24}$. We use this upper bound for $\psi(\Sigma)$ and lower bound for $\psi(X)$ and $\psi(Y)$. The duplication formula [21, 5.5.5] gives $2\psi(2y) = \psi(y) + \psi(y + \frac{1}{2}) + 2\log 2$, and ψ is increasing, so $2\psi(N+2\tau) - 2\psi(h+\tau) > 2\log 2$. Hence the derivative is less than $\log \frac{\Sigma^2}{4XY} + (\frac{1}{2X} + \frac{1}{2Y} - \frac{1}{\Sigma}) + \frac{1}{12}(\frac{1}{X^2} + \frac{1}{Y^2})$. Now $4XY = \Sigma^2(1-z)$ and $X^2 + Y^2 = \Sigma^2(1+z)/2$, so the 3 terms are $-\log(1-z)$, $\frac{X^2+Y^2}{2XY\Sigma} = \frac{1+z}{(1-z)\Sigma}$ and $\frac{X^2+Y^2}{12X^2Y^2} = \frac{2(1+z)}{3\Sigma^2(1-z)^2}$. $\square$

Proposition A.4. Let $0 < \varepsilon < \frac{1}{4}$, $\kappa = \frac{2\varepsilon}{1-4\varepsilon}$, $\kappa_s = \kappa(s+10)$ and $\Sigma_s = s + 2\kappa_s$. For $1 \leq s \leq N-1$ and $b \in \mathcal{S}_s$ let $\Delta_s(b) = \mathcal{L}_s \Phi_{s+1}^{\kappa_{s+1}}(b) / \Phi_s^{\kappa_s}(b)$.

(a) Let $c = \kappa_s$, $c' = \kappa_{s+1}$, $d = \frac{s}{2} - b$, $u = \frac{d}{s+2c'}$, $v = \frac{d}{N-s}$ and $\rho = \frac{s+2c'}{s(1+2\beta)}$. Then $|u| < \frac{1}{2}$ and

$$\Delta_s(b) = \mathcal{J} \frac{\Phi_s^{c'}(b)}{\Phi_s^c(b)}, \quad \mathcal{J} = 1 - (\rho - 1) \frac{u(u+v)}{\frac{1}{4} - u^2}.$$

(b) $\log \Delta_s(b) \leq \frac{\kappa}{\Sigma_s} + \frac{2\kappa}{3\Sigma_s^2}$.

(c) For $1 \leq t \leq h$, $V_{t+1}(1) \leq \pi_t \exp(\kappa[H_{N-1} - H_t + \frac{2}{3t}] + \kappa_{t+1}[\log \frac{(t+1)^2}{4t} + \frac{2}{3}])$.

(d) $C_N \leq U_N := \varepsilon c_1 e^{\kappa H_{N-1}} \sum_{t=1}^h w_t (1-\varepsilon)^{t-1} e^{\kappa \mathcal{G}(t)}$, where $\mathcal{G}(t) = (t+11)(\log \frac{(t+1)^2}{4t} + \frac{2}{3}) - H_t + \frac{2}{3t}$.

The slope κ solves $2(\kappa - \beta)/(1 + 2\beta) = \kappa$, so that in (b) the gain in $\mathcal{J}$ cancels the main loss from raising the extra balls from c to c' .

Proof. (a) Write $r' = r_s^{c'}(b)$, $\bar{r} = \bar{r}_s(b)$ and $r = r_s(b)$. By Lemma A.1(b) with c' , and the recursion in Lemma A.1(a), we have $r' \Phi_{s+1}^{c'}(b+1) = \bar{r} \Phi_s^{c'}(b)$ and $(1-r') \Phi_{s+1}^{c'}(b) = (1-\bar{r}) \Phi_s^{c'}(b)$. So the numerator of $\Delta_s(b)$ divided by $\Phi_s^{c'}(b)$ is $\frac{(1-r')(1-\bar{r})}{1-r'} + \frac{r\bar{r}}{r'}$. Here $r' = \frac{1}{2} - u$, $\bar{r} = \frac{1}{2} + v$ and $r = \frac{1}{2} - \rho u$, since $\frac{1}{2} - r = (1-2\varepsilon)\frac{d}{s} = \rho u$. Over the denominator $r'(1-r') = \frac{1}{4} - u^2$ the numerator is $f(u, v) + f(-u, -v)$ with $f(u, v) = (\frac{1}{2} + \rho u)(\frac{1}{2} - v)(\frac{1}{2} - u)$, and this equals $\frac{1}{4} - \rho u^2 - (\rho - 1)uv$. Also $|d| \leq \frac{s}{2} - 1$, so $|u| < \frac{1}{2}$.

(b) *Step 1: the factor $\mathcal{J}$.* Write $c = \kappa_s$, $c' = c + \kappa$ and $\Sigma = \Sigma_s$. Since $c' - \beta s = (\kappa - \beta)s + 11\kappa$, the choice of κ gives $\rho - 1 = \kappa + \nu_s$ with $\nu_s = \frac{22\kappa}{s(1+2\beta)}$. As u and v have the sign of d , $u(u+v) \geq u^2$. So $\mathcal{J} \leq 1 - (\kappa + \nu_s)W'$ with $z' = 4u^2$ and $W' = \frac{z'}{1-z'}$. Also $\mathcal{J} > 0$ by (a), since $0 < r < 1$ and $\bar{r}$, $1 - \bar{r}$ are not both 0. So $\log \mathcal{J} \leq -(\kappa + \nu_s)W'$.

Step 2: the ratio $\Phi_s^{c'}/\Phi_s^c$. The logarithm $\log(\Phi_s^{c'}(b)/\Phi_s^c(b))$ is the integral over $\tau \in [c, c']$ of the derivative in Lemma A.3. Its bound increases in z and decreases in Σ , while z decreases and Σ increases in τ . So we may take the bound at $\tau = c$, where $X + Y = \Sigma$, and multiply it by the length κ of the interval. Put $W = \frac{z}{1-z}$. Then $\frac{1}{1-z} = 1 + W$, $\frac{1+z}{1-z} = 1 + 2W$ and $-\log(1-z) \leq W$, so

$$\log \Delta_s(b) \leq \kappa \left[W + \frac{1+2W}{\Sigma} + \frac{2(1+2W)(1+W)}{3\Sigma^2} \right] - (\kappa + \nu_s)W'. \quad (2)$$

Step 3: 3 facts. (i) $X, Y \geq 1 + c$, since $b \geq 1$ and $s - b \geq 1$. So $XY \geq (1+c)\Sigma/2$ and $1 + W = \frac{\Sigma^2}{4XY} \leq \frac{\Sigma}{2(1+c)}$. (ii) $z' = \lambda z$ with $\lambda = (\frac{\Sigma}{\Sigma+2\kappa})^2 \geq 1 - \frac{4\kappa}{\Sigma}$. So $W - W' = \frac{(1-\lambda)z}{(1-z)(1-\lambda z)} \leq (1-\lambda)W(1+W) \leq \frac{2\kappa}{1+c}W$ by (i). As $c \geq 11\kappa$, this gives $W' \geq \frac{9}{11}W$. (iii) $\Sigma \geq s$ and $1 + c > \kappa_s$. If $W > 0$ then $s \neq 2b$, and with $b \geq 1$, $s - b \geq 1$ this forces $s \geq 3$, so $\Sigma^2 \geq 3s$.

Step 4: collect the terms. By (ii) and (iii), $\kappa(W - W') \leq \frac{2\kappa^2}{1+c}W \leq \frac{2\kappa}{s}W$ and $\frac{2\kappa W}{\Sigma} \leq \frac{2\kappa}{s}W$. Since $(1 + 2W)(1 + W) = 1 + 3W + 2W^2$, κ times the last term in the bracket of (2) is $\frac{2\kappa}{3\Sigma^2} + \frac{2\kappa W}{\Sigma^2} + \frac{4\kappa W^2}{3\Sigma^2}$. Here $\frac{2\kappa W}{\Sigma^2} \leq \frac{2\kappa}{3s}W$ by (iii), and $\frac{4\kappa W^2}{3\Sigma^2} \leq \frac{4\kappa W(1+W)}{3\Sigma^2} \leq \frac{2\kappa W}{3\Sigma(1+c)} \leq \frac{2\kappa}{3s}W$ by (i). With $-\nu_s W' \leq -\frac{9}{11}\nu_s W$ this gives

$$\log \Delta_s(b) \leq \frac{\kappa}{\Sigma} + \frac{2\kappa}{3\Sigma^2} + W \left[\frac{16\kappa}{3s} - \frac{18\kappa}{s(1+2\beta)} \right],$$

and the bracket is negative because $1 + 2\beta < 2$ for $\varepsilon < \frac{1}{4}$, so $18/(1+2\beta) > 9 > 16/3$.

(c) Apply Lemma A.2(a) with $s_0 = t + 1$, $F_s = \Phi_s^{\kappa_s}$ and $K_s = \exp(\frac{\kappa}{s} + \frac{2\kappa}{3s^2})$. This is allowed, since $F_N(h) = 1$, $F_s > 0$ on $\mathcal{S}_s$, and $\mathcal{L}_s F_{s+1} = \Delta_s F_s \leq K_s F_s$ on $\mathcal{S}_s$ by (b) and $\Sigma_s \geq s$. Since $1 \in \mathcal{S}_{t+1}$ and $\sum_{s>t} s^{-2} \leq \frac{1}{t}$, we get $V_{t+1}(1) \leq e^{\kappa[H_{N-1}-H_t+2/(3t)]} \Phi_{t+1}^{\kappa_{t+1}}(1)$. Now $\log(\Phi_{t+1}^{\kappa_{t+1}}(1)/\pi_t)$ is the integral over $\tau \in [0, \kappa_{t+1}]$ of the derivative in Lemma A.3, whose bound is largest at $\tau = 0$. There $X = 1$, $Y = t$ and $\frac{1}{1-z} = \frac{(t+1)^2}{4t}$, and the bound is $\log \frac{(t+1)^2}{4t} + \frac{t^2+1}{2t(t+1)} + \frac{t^2+1}{12t^2} \leq \log \frac{(t+1)^2}{4t} + \frac{2}{3}$.

(d) Insert (c) into Proposition 2.5(a), and use $\kappa_{t+1} = \kappa(t+1)$ and $\pi_t = c_1 w_t$. $\square$

Proof of Theorem 2.6. (a) By Proposition 2.5(a) and Lemma A.1(d), $C_N \geq \varepsilon c_1 \sum_t w_t (1 - \varepsilon)^{t-1}$. Now $(1 - \varepsilon)^{t-1} \geq 1 - (t-1)\varepsilon$ and $\sum_t w_t (t-1) = 2 - \frac{6}{h+2}$ by Proposition 2.5(c). So $C_N \geq \varepsilon c_1 (1 - 2\varepsilon)$, and $c_1 = \frac{4}{N+2}$.

(b) *Step 1: the constants.* Let $N \geq 64$ and $\varepsilon(1 + \log N) \leq \frac{1}{100}$. Then $\varepsilon < 0.002$ and $\kappa \leq 2\varepsilon(1 + 8\varepsilon) \leq 2.1\varepsilon$. Since $\log \frac{(t+1)^2}{4t} \geq 0$ and $\frac{2}{3}(t+1) > 1 + \log t \geq H_t$, we have $\mathcal{G}(t) > 0$. Since $\frac{(t+1)^2}{4t} \leq \frac{t+1}{2}$, $\frac{2}{3} < \log 2$ and $H_t \geq \frac{2}{3t}$, also $\mathcal{G}(t) \leq (t+1) \log(t+1) \leq 12t \log(t+1)$. So $\kappa \mathcal{G}(t) \leq 25.2\varepsilon t \log(t+1)$, and this is at most $\frac{t}{2} \log 2$ for $t \leq h$, since then $25.2\varepsilon \log(t+1) \leq 25.2\varepsilon \log N \leq 0.252 < \frac{1}{2} \log 2$.

Step 2: the sum over t. Since $e^y - 1 \leq ye^y$ for $y \geq 0$, Step 1 gives $e^{\kappa \mathcal{G}(t)} - 1 \leq \kappa \mathcal{G}(t) 2^{t/2}$. By Proposition A.4(d), $(1 - \varepsilon)^{t-1} \leq 1$, $\sum_t w_t = 1$ and $w_t \leq 2t2^{-t}$ (Proposition 2.5(c)),

$$C_N \leq \varepsilon c_1 e^{\kappa H_{N-1}} \left(1 + 50.4\varepsilon \sum_{t \geq 1} t^2 \log(t+1) 2^{-t/2} \right),$$

and we bound the series as follows. The ratio of its consecutive terms is $(1 + \frac{1}{t})^2 \frac{\log(t+2)}{\log(t+1)} 2^{-1/2}$, which decreases in t and is below $\frac{3}{4}$ at $t = 60$. So the terms with $t > 60$ add up to less than 4 times the term at $t = 61$, which is below 10^{-4} . The first 60 terms add up to 102.5864 to 4 decimals (computed in 30-digit arithmetic by `verify_elephant_crossing.py`), so the series is below 102.59.

Step 3: the factor $e^{\kappa H_{N-1}}$. Finally $H_{N-1} \leq 1 + \log N$, so $\kappa H_{N-1} \leq 2\varepsilon \log N + 2\varepsilon + 16\varepsilon^2(1 + \log N) \leq 2\varepsilon \log N + 3\varepsilon$. With $1 + y \leq e^y$ this proves (b), since $3 + 50.4 \cdot 102.59 < 5200$. $\square$

B Proofs for Section 3

We keep the conventions of Section 3. We write C for an absolute constant whose value may change from line to line, and $\bar{c}$, A_0 , A_1 for fixed absolute constants. Since M is independent of X_1 , $f_N(m) = \mathbb{P}(M_N = m)$. Every path of a chain $M^{[k_0]}$ is nondecreasing in time, and we use the paths of Y and $Z^{(j)}$ fixed after Lemma 3.3. The family $\mathcal{T}_v$ has the law β_v of Lemma 2.4. Let $\bar{\beta}_v(s) = \mathbb{P}(|\mathcal{T}_v| \geq s)$, with $\bar{\beta}_v(s) = 1$ for $s \leq 1$. By Pascal's rule $\binom{N-s}{v-1} - \binom{N-s-1}{v-1} = \binom{N-s-1}{v-2}$ the

tail of β_v telescopes. So, for $1 \leq s \leq N-1$,

$$\bar{\beta}_v(s) = \binom{N-s}{v-1} \bigg/ \binom{N-1}{v-1} = \prod_{i=1}^{v-1} \left(1 - \frac{s-1}{N-i}\right) \leq e^{-(v-1)(s-1)/(N-1)}, \quad \beta_v(s) = \frac{v-1}{N-s} \bar{\beta}_v(s), \quad (3)$$

where the product vanishes as soon as one factor does. Where they are defined, the ratios are

$$\frac{\beta_v(d)}{\beta_v(d-1)} = 1 - \frac{v-2}{N-d}, \quad \frac{\bar{\beta}_v(s-1)}{\bar{\beta}_v(s)} = 1 + \frac{v-1}{N-s-v+2}. \quad (4)$$

Lemma B.1 (monotone sandwich). *Let $0 \leq a \leq b \leq N$. If $m_h \leq a$, then $(b-a+1)f_N(b) \leq \mathbb{P}(a \leq M_N \leq b)$. If $b \leq m_h$, then $(b-a+1)f_N(a) \leq \mathbb{P}(a \leq M_N \leq b)$.*

Proof. By Lemma 3.4, $f_N(\ell) \geq f_N(b)$ for $a \leq \ell \leq b$ in the first case, and $f_N(\ell) \geq f_N(a)$ in the second. Sum over ℓ . $\square$

Lemma B.2 (tail). *Let $0 < \varepsilon < 1$, $k_0 \geq 1$ and $t \geq k_0$. For every integer $z \geq 1$,*

$$\mathbb{P}(M_t^{[k_0]} \geq z) \leq \frac{\varepsilon t H_z}{z}, \quad \mathbb{E}M_t^{[k_0]} \leq \varepsilon t \log \frac{t}{k_0}.$$

In particular $\mathbb{P}(M_N \geq z) \leq xH_z/z$, and $\mathbb{P}(Z_s^{(j)} \geq z) \leq xH_z/z$ for $j-1 \leq s \leq N$.

Proof. We use the tree of Lemma 3.1 on $[t]$, with marks only on $v > k_0$. Every odd vertex lies in the family of a marked vertex, so $M_t^{[k_0]} \leq \sum_{v \text{ marked}} |\mathcal{T}_v|$. So $M_t^{[k_0]} \geq z$ needs a marked family of size at least z , or $\sum_{v \text{ marked}} |\mathcal{T}_v| \mathbf{1}\{|\mathcal{T}_v| < z\} \geq z$. By the union bound and Markov's inequality, and since marks and tree are independent, the probability is at most $\varepsilon \sum_v \mathbb{P}(|\mathcal{T}_v| \geq z) + \frac{\varepsilon}{z} \sum_v \mathbb{E}[|\mathcal{T}_v|; |\mathcal{T}_v| < z]$. By the first sum in Lemma 2.4, with t in place of N , this is at most $\varepsilon \frac{t}{z} + \frac{\varepsilon}{z} \sum_{d=1}^{z-1} \frac{t}{d+1} = \frac{\varepsilon t H_z}{z}$. For the mean, $\mathbb{E}M_t^{[k_0]} \leq \varepsilon \sum_{v=k_0+1}^t \frac{t}{v} \leq \varepsilon t \log \frac{t}{k_0}$. $\square$

Lemma B.3 (late start). *Let $j \geq 2$, $t \geq j-1$ and $z \geq 0$. Then $\mathbb{P}(M_t^{[j-1]} \geq z) \leq \exp\{-\frac{j-1}{2}(\frac{z}{t} - \varepsilon \log \frac{2t}{j-1})\}$.*

Proof. Write $G_k = M_k^{[j-1]}$ and $\nu = (j-1)/2$. For $k > \nu$ let $\theta_k = \log \frac{k}{k-\nu}$, so that $e^{\theta_k} - 1 = \frac{\nu}{k-\nu}$. Since $r_k(c) \leq \varepsilon + c/k$ and $1+\nu \leq e^\nu$, $\mathbb{E}[e^{\theta_{k+1}G_{k+1}} | G_k = c] \leq \exp\{c(\theta_{k+1} + \frac{\nu}{k(k+1-\nu)}) + \frac{\varepsilon\nu}{k+1-\nu}\}$. Now $\theta_k - \theta_{k+1} = \log(1+w)$ with $w = \frac{\nu}{(k-\nu)(k+1)}$, and $\log(1+w) \geq \frac{w}{1+w} = \frac{\nu}{k(k+1-\nu)}$. So $\mathbb{E}e^{\theta_{k+1}G_{k+1}} \leq e^{\varepsilon\nu/(k+1-\nu)} \mathbb{E}e^{\theta_k G_k}$ for $k \geq j-1$. Starting from $G_{j-1} = 0$ and using $\sum_{k=j}^t \frac{1}{k-\nu} \leq \log \frac{t-\nu}{\nu} \leq \log \frac{2t}{j-1}$, we get $\mathbb{E}e^{\theta_t G_t} \leq \exp\{\varepsilon\nu \log \frac{2t}{j-1}\}$. Markov's inequality with $\theta_t \geq \nu/t$ gives the claim. $\square$

Lemma B.4 (the first flipped family). *For $1 \leq d \leq N-1$,*

$$\frac{x}{d(d+1)} \left(1 - \frac{2x(N-d-1)}{N(d+2)}\right) \leq \mathbb{P}(D = d) \leq \frac{x}{d(d+1)}, \quad \mathbb{P}(D \geq d) \leq x \left(\frac{1}{d} - \frac{1}{N}\right).$$

Proof. By Lemma 3.3, $\mathbb{P}(D = d) = \sum_j \varepsilon(1-\varepsilon)^{j-2} \beta_j(d)$, and $1 - (j-2)\varepsilon \leq (1-\varepsilon)^{j-2} \leq 1$. Apply the 2 sums of Lemma 2.4, and sum the upper bound over $d' \geq d$. $\square$

Proposition B.5 (local lower bound). *There is an absolute constant C such that the following hold.*

(a) For $2 \leq m \leq N/4$, $f_N(m) \geq \frac{x}{m(m+1)}(1 - e_1(m))$ with $e_1(m) = C\left(\frac{x \log(m+1)}{m} + \varepsilon \log N + me^{-m/9}\right)$.

(b) For $N/4 < m \leq N - 16$ and $r = N - m$, $f_N(m) \geq \frac{x}{m(m+1)}(1 - e_2(m))$ with $e_2(m) = C\left(\frac{x \log N}{r} + \frac{x \log N}{m} + Ne^{-r/40}\right)$.

Proof. (a) *Step 1: one value of j .* Fix $2 \leq j \leq J := \lfloor N/8 \rfloor$ and put $\Gamma_d = d - Y_d + Z_{N-d}^{(j)}$. The inner sum in Lemma 3.3 is $Q_j(m) = \mathbb{E} \sum_d \beta_j(d) \mathbf{1}\{\Gamma_d = m\}$, and every step of Γ lies in $\{-1, 0, 1\}$. On the event $\{Z_{N-1}^{(j)} \leq m - 1, Y_{2m} \leq m\}$ we have $\Gamma_1 = 1 + Z_{N-1}^{(j)} \leq m$, and at $d^\circ = m + Y_{2m} \leq 2m$ we have $\Gamma_{d^\circ} \geq d^\circ - Y_{2m} = m$, since Y is nondecreasing and $Z \geq 0$ (and $d^\circ \leq N - j + 1$ because $2m + j \leq N$). So Γ hits m at some $d \leq d^\circ$, and as β_j is nonincreasing the sum is at least $\beta_j(m + Y_{2m})$. By (4), $\beta_j(m + y)/\beta_j(m) = \prod_{i=1}^y (1 - \frac{j-2}{N-m-i}) \geq 1 - \frac{2y(j-2)}{N}$ for $y \leq m$, since each fraction is at most $\frac{2(j-2)}{N} \leq \frac{1}{4}$. By independence and $ab \geq a + b - 1$ for $a, b \in [0, 1]$, $Q_j(m) \geq \beta_j(m)[1 - \mathbb{P}(Y_{2m} > m) - \mathbb{P}(Z_{N-1}^{(j)} \geq m)] - \frac{2(j-2)}{N} \mathbb{E} Y_{2m} \beta_j(m)$.

Step 2: sum over j . Lemma B.2 gives $\mathbb{P}(Y_{2m} > m) \leq 2\varepsilon H_{m+1}$, $\mathbb{P}(Z_{N-1}^{(j)} \geq m) \leq xH_m/m$ and $\mathbb{E} Y_{2m} \leq 2\varepsilon m \log N$. If $2\varepsilon H_{m+1} + xH_m/m > 1$, then $e_1(m) > 1$ and there is nothing to prove. Otherwise we multiply by $\mathbb{P}(V = j) \leq \varepsilon$ and sum over $2 \leq j \leq J$. By the second sum in Lemma 2.4, the terms with $\mathbb{E} Y_{2m}$ add up to at most $\frac{4\varepsilon m \log N}{N} \cdot \frac{2x(N-m-1)}{m(m+1)(m+2)} \leq \frac{x}{m(m+1)} 8\varepsilon \log N$. By Lemma B.4, $\sum_{j \leq J} \mathbb{P}(V = j) \beta_j(m) \geq \frac{x}{m(m+1)}(1 - \frac{2x}{m+2}) - \varepsilon \sum_{j > J} \beta_j(m)$. *Step 3: the late first flips $j > J$.* By (3), $\beta_j(m) \leq \frac{j-1}{N-m} e^{-(j-1)a}$ with $a = \frac{m-1}{N-1} \leq \frac{1}{4}$, and $aJ \geq \frac{m-1}{9}$ because $\lfloor N/8 \rfloor \geq \frac{N-1}{9}$ for $N \geq 64$. Summing the series, with $1 - e^{-a} \geq a/2$ and $N - m \geq \frac{3}{4}N$, gives $\varepsilon \sum_{j > J} \beta_j(m) \leq \frac{C\varepsilon}{N}(\frac{J}{a} + \frac{1}{a^2})e^{-(m-1)/9} \leq \frac{x}{m(m+1)} Cme^{-m/9}$. Collecting the terms gives (a).

(b) The same argument works with $J = w = \lfloor r/8 \rfloor \geq 2$, the event $\{Z_{N-1}^{(j)} \leq m - 1, Y_{m+w} \leq w\}$ and $d^\circ = m + Y_{m+w} \leq m + w$. Note that $m + w + j \leq N$. Only 3 estimates change. For $y \leq w$ and $j \leq J$ we have $N - m - i \geq \frac{7}{8}r$, so $\beta_j(m + y) \geq \beta_j(m)(1 - \frac{8y(j-2)}{7r})$. Since $w + 1 \geq r/8$, Lemma B.2 gives $\mathbb{P}(Y_{m+w} > w) \leq 8xH_N/r$ and $\mathbb{E} Y_{m+w} \leq x \log N$, and the terms with $\mathbb{E} Y_{m+w}$ add up to at most $\frac{x}{m(m+1)} \cdot \frac{Cx \log N}{m}$. For the dropped terms, $a \geq \frac{1}{5}$ and $J \geq \frac{r-7}{8}$, so $\varepsilon \sum_{j > J} \beta_j(m) \leq \frac{C\varepsilon(J+1)}{r} e^{-J/5} \leq \frac{x}{m(m+1)} CNe^{-r/40}$. $\square$

Proposition B.6 (cumulative upper bound). *There is an absolute constant C such that for $6 \leq m \leq N/2$ with $m \geq 16x \log m$ we have $\mathbb{P}(M_N \geq m) \leq \frac{x}{m}(1 + C \frac{x \log^2 m}{m})$.*

Proof. *Step 1: split by the first flip.* Let $J = \lfloor N/4 \rfloor$ and $Z_j = Z_{N-1}^{(j)}$. By Lemma 3.3, given $V = j$ the count M_N has the law of $D - Y_D + Z_{N-D}^{(j)} \leq D + Z_j$, with D of law β_j independent of Z_j . On $\{V > J\}$ there is no mark on $2, \dots, J$, so by Lemma 3.1 M_N is a copy of $M_N^{[J]}$ independent of $\{V > J\}$. For $0 \leq z \leq m/8$, (4) gives $\bar{\beta}_j(m - z) \leq \bar{\beta}_j(m) \varrho_j^z$ with $\varrho_j = 1 + \frac{j-1}{N-m-j+2} \leq 1 + \frac{4(j-1)}{N}$, and $\varrho^z - 1 \leq z(\varrho - 1)\varrho^z$. Hence

$$\mathbb{P}(M_N \geq m) \leq \sum_{j=2}^J \mathbb{P}(V = j) \left[\bar{\beta}_j(m) (1 + (\varrho_j - 1) \varrho_j^{m/8} \mathbb{E} Z_j) + \mathbb{P}\left(Z_j > \frac{m}{8}\right) \right] + \mathbb{P}(V > J) \mathbb{P}(M_N^{[J]} \geq m).$$

This gives 4 terms, which we bound in turn. The main term is $\sum_j \mathbb{P}(V = j) \bar{\beta}_j(m) \leq \mathbb{P}(D \geq m) \leq \frac{x}{m}$ by Lemma B.4.

Step 2: the shift term. We use $\mathbb{P}(V = j) \leq \varepsilon$, $\bar{\beta}_j(m) \leq e^{-(j-1)(m-1)/(N-1)}$, $\varrho_j^{m/8} \leq e^{(j-1)m/(2N)}$ and $\mathbb{E}Z_j \leq x \log \frac{N}{j-1}$ (Lemma B.2). As $m \geq 6$, the exponents add up to at most $-(j-1)a$ with $a = \frac{m}{3N}$. With $i = j-1$ and $\log \frac{N}{i} = \log \frac{m}{3} + \log \frac{1}{ai}$, the shift term is at most $\frac{4\varepsilon x}{N} [\log \frac{m}{3} \sum_{i \geq 1} i e^{-ai} + \sum_{i \leq 1/a} i \log \frac{1}{ai}] \leq \frac{4\varepsilon x}{N} \cdot \frac{\log m + C}{a^2} \leq \frac{x}{m} \cdot \frac{Cx \log^2 m}{m}$, since $\sum_i i e^{-ai} \leq a^{-2}$ and $t \log \frac{1}{at} \leq \frac{1}{ea}$ for $0 < t \leq 1/a$.

Step 3: the big-Z term. Let $\mathcal{K} = 32 \log(m/x)$ and $j_{\mathcal{K}} = \lfloor \mathcal{K}N/m \rfloor + 1$. Note that $m \geq 16x \log 6 > 28x$, so $\mathcal{K} \geq 2$. For $j \leq j_{\mathcal{K}}$, Lemma B.2 gives $\mathbb{P}(Z_j > m/8) \leq 8xH_m/m$, and these j contribute at most $\varepsilon \frac{\mathcal{K}N}{m} \cdot \frac{8xH_m}{m} \leq \frac{x}{m} \cdot \frac{Cx \log^2 m}{m}$. For $j > j_{\mathcal{K}}$ we have $\frac{2(N-1)}{j-1} \leq \frac{2m}{\mathcal{K}} \leq m$, so $\varepsilon \log \frac{2(N-1)}{j-1} \leq \frac{x \log m}{N} \leq \frac{m}{16N}$, and Lemma B.3 with $t = N-1$ gives $\mathbb{P}(Z_j \geq m/8) \leq e^{-(j-1)m/(32N)}$. Summing over $j-1 \geq j_{\mathcal{K}} > \mathcal{K}N/m$ gives at most $\varepsilon \frac{e^{-\mathcal{K}/32}}{1-e^{-m/(32N)}} \leq \varepsilon \left(\frac{32N}{m} + 1 \right) \frac{x}{m} \leq \frac{x}{m} \cdot \frac{33x}{m}$.

Step 4: no early flip. Finally $\mathbb{P}(V > J) = (1 - \varepsilon)^{J-1} \leq e^{\varepsilon-3x/16}$, since $J \geq 3N/16$, and Lemma B.3 with $j-1 = J \in [\frac{3N}{16}, \frac{N}{4}]$ and $t = N$ gives $\mathbb{P}(M_N^{[J]} \geq m) \leq \exp\{-\frac{3m}{32} + \frac{x}{8} \log \frac{32}{3}\}$. As $x \leq m/28$, the last term is at most $e^{1/4-m/13} \leq \frac{x}{m} \cdot \frac{Cx \log^2 m}{m}$. $\square$

Lemma B.7 (mode bound). *There are absolute constants $\bar{c} > 0$ and A_0 such that if $\varepsilon \log N \leq \bar{c}$ and $A_0 x \log(x+2) \leq N/4$, then $m_h \leq A_0 x \log^2(x+1)$.*

Proof. Let $\ell = \lceil Ax \log(x+2) \rceil$. Since $\log(\ell+1) \leq \log(A+2) + 2 \log(x+2)$, we have $\frac{x \log(\ell+1)}{\ell} \leq \frac{\log(A+2)+2}{A}$, and $\ell e^{-\ell/9}$ is small when A is large. So we can fix the absolute constants A and $\bar{c}$ such that $e_1(\ell) \leq \frac{1}{2}$ in Proposition B.5(a) whenever $\varepsilon \log N \leq \bar{c}$. We take $A_0 \geq A+1$, so that $2 \leq \ell \leq N/4$. Then $f_N(\ell) \geq \frac{x}{2\ell(\ell+1)}$. If $m_h \geq \ell$, Lemmas B.1 (with $a = \ell$, $b = m_h$) and B.2 give $(m_h - \ell + 1)f_N(\ell) \leq \mathbb{P}(M_N \geq \ell) \leq xH_\ell/\ell$. So $m_h \leq (\ell+1)(1+2H_\ell)$ in every case. As $\log \ell \leq C \log(x+2)$ and $\log(x+2) \leq 2 \log(x+1)$, this is at most $A_0 x \log^2(x+1)$ once A_0 is large. $\square$

Proposition B.8 (local upper bound). *There are absolute constants A_1 and C such that if $\varepsilon \log N \leq \bar{c}$ and $A_1 x \log^2(x+1) \leq m \leq N/8$, then*

$$f_N(m) \leq \frac{x}{m(m+1)} (1 + C\Delta(m)^{1/2}), \quad \Delta(m) = \frac{x \log^2 m}{m} + \varepsilon \log N + \frac{m}{N}.$$

Proof. We take $A_1 \geq 2A_0$ so large that Lemma B.7 applies and every $m' \geq m/2$ satisfies $m' \geq 6$ and $m' \geq 16x \log m'$ (as $t/\log t$ increases for $t \geq e$). *Step 1: a difference of tails.* Let $1 \leq k \leq m/2$. Then $m-k \geq m/2 \geq m_h$, and Lemma B.1 with $a = m-k$ and $b = m$ gives $(k+1)f_N(m) \leq \mathbb{P}(M_N \geq m-k) - \mathbb{P}(M_N \geq m+1)$. *Step 2: bound each tail.* Proposition B.6 at $m-k$ bounds the first term by $\frac{x}{m-k} + \frac{Cx^2 \log^2 m}{m^2}$. For the second we sum Proposition B.5(a) over $m < \ell \leq N/4$. Since $\sum_{m < \ell \leq N/4} \frac{1}{\ell(\ell+1)} \geq \frac{1}{m+1} - \frac{4}{N}$, $\frac{\log(\ell+1)}{\ell}$ and $\ell e^{-\ell/9}$ decrease for $\ell \geq 9$, and $m e^{-m/9} \leq C/m \leq C\Delta(m)$, we get $\mathbb{P}(M_N \geq m+1) \geq \frac{x}{m+1} (1 - C\Delta(m))$. As $\frac{x}{m-k} - \frac{x}{m+1} = \frac{x(k+1)}{(m-k)(m+1)}$, this gives $f_N(m) \leq \frac{x}{(m-k)(m+1)} + \frac{Cx\Delta(m)}{(k+1)m}$. *Step 3: choose k .* If $\Delta(m) \leq \frac{1}{16}$, take $k = \lceil m\Delta(m)^{1/2} \rceil \leq \frac{m}{2}$. Then $\frac{m}{m-k} \leq 1 + \frac{2k}{m} \leq 1 + 2\Delta^{1/2} + \frac{2}{m}$ and $\frac{\Delta}{k+1} \leq \frac{\Delta^{1/2}}{m}$, which gives the claim, as $\frac{1}{m} \leq \Delta(m) \leq \Delta(m)^{1/2}$. If $\Delta(m) > \frac{1}{16}$, take $k = \lfloor m/2 \rfloor$. Then $f_N(m) \leq \frac{x}{m(m+1)} (2 + C\Delta(m))$, and the claim follows after enlarging C , since $\Delta(m)$ is bounded by an absolute constant ($\frac{x \log^2 m}{m}$ decreases for $m \geq e^2$ and is bounded at $m = A_1 x \log^2(x+1)$). $\square$

Proof of Theorem 3.8. Step 1: the hypotheses hold. Put $\ell_x = \log(x+1) \geq \log 2$ and let $\mathcal{R}_N$ be the range $\omega_N x \ell_x^2 \leq m \leq \delta'_N N$. Since $x \log^2 N/N \rightarrow 0$, for large N we have $\varepsilon(1 + \log N) \leq \min(\bar{c}, \frac{1}{100})$,

$\omega_N \geq \max(A_1, 16)$, $\delta'_N \leq \frac{1}{8}$ and $A_0 x \log(x+2) \leq N/4$. *Step 2: the errors are small on $\mathcal{R}_N$.* The function $\frac{x \log^2 m}{m}$ decreases for $m \geq e^2$. At $m = \omega_N x \ell_x^2$ we have $m \geq e^2$ and $\log m \leq \log \omega_N + 3\ell_x$, as $\log x$ and $\log \ell_x$ are at most ℓ_x . So on $\mathcal{R}_N$, uniformly in $x \geq 1$, $\frac{x \log^2 m}{m} \leq \frac{(\log \omega_N + 3\ell_x)^2}{\omega_N \ell_x^2} \leq \frac{1}{\omega_N} (\frac{\log \omega_N}{\log 2} + 3)^2 \rightarrow 0$. Also $m \geq \omega_N \log^2 2 \rightarrow \infty$ on $\mathcal{R}_N$. Since $\log(m+1) \leq 2 \log^2 m$ for $m \geq 3$, Proposition B.5(a) has $e_1(m) \leq C(\frac{x \log^2 m}{m} + \varepsilon \log N + m e^{-m/9}) \rightarrow 0$ uniformly on $\mathcal{R}_N$. Also $\Delta(m) \leq \frac{x \log^2 m}{m} + \varepsilon \log N + \delta'_N \rightarrow 0$ there, and Proposition B.8 applies. So $f_N(m) = \frac{x}{m(m+1)}(1+o(1))$ uniformly on $\mathcal{R}_N$.

Step 3: the mirror term. Lemma B.7 gives $m_h \leq A_0 x \ell_x^2 \leq h$ for large N , since $x \ell_x^2 \leq x \log^2(N+1) = o(N)$. As $N-m \geq h$, Lemma 3.4 gives $0 \leq f_N(N-m) \leq f_N(h) = C_N$. By Theorem 2.6(b) and $\varepsilon \log N \rightarrow 0$, $C_N \leq \frac{4x}{N(N+2)} N^{2\varepsilon} e^{5200\varepsilon} = \frac{4x}{N^2} (1+o(1))$. On $\mathcal{R}_N$ this is at most $\frac{x}{m(m+1)} \cdot 8\delta_N'^2 (1+o(1))$, since $m(m+1) \leq 2\delta_N'^2 N^2$. Hence $P_N(N-m) = \frac{1}{2}[f_N(m) + f_N(N-m)] = \frac{x}{2m(m+1)}(1+o(1))$ uniformly on $\mathcal{R}_N$, and $m(m+1) = m^2(1+o(1))$ because $m \rightarrow \infty$. $\square$

C Proof of Theorem 4.4

We use the notation of Section 4, with $1 < \alpha < 2$, $a_m = m^{1/\alpha}$ and $\tau_0 = 0$. For a first run D let $\tau_m^D = D + \tau'_{m-1}$, where (τ'_m) is an independent copy of (τ_m) , and $\tau_0^D = 0$. Then $u_m^D(k) = \mathbb{P}(\tau_m^D = k)$ and $q_m^D(k) = \mathbb{P}(\tau_{m-1}^D < k \leq \tau_m^D)$. Conditioning on the last run gives, for $m, k \geq 1$,

$$\begin{aligned} q_m(k) &= \sum_{j \geq 1} \bar{F}(j) u_{m-1}(k-j), & q_{m+1}^D(k) &= \sum_{j \geq 1} \bar{F}(j) u_m^D(k-j), \\ u_m^D(k) &= \sum_{d \geq 1} \mathbb{P}(D = d) u_{m-1}(k-d). \end{aligned} \tag{5}$$

Since τ_m and τ_m^D increase strictly, $\sum_m q_m(k) = \sum_m q_m^D(k) = 1$, $\sum_m u_m^D(k) \leq 1$ and $\sum_{m \leq m'} q_m(k) = \mathbb{P}(\tau_{m'} \geq k)$. We put $z_{m,k} = (k - m\mu)/a_m$ and $\delta_m = \sup_k |a_m u_m(k) - g(z_{m,k})|$, so that $\delta_m \rightarrow 0$ by Lemma 4.3(iii). Let ω be the modulus of continuity of g and $\bar{g}(x) = \sup_{|y| \geq x} g(y)$. By Lemma 4.3(ii), $\omega(\eta) \rightarrow 0$ as $\eta \rightarrow 0$ and $\bar{g}(x) \rightarrow 0$ as $x \rightarrow \infty$.

Lemma C.1. *There is a constant $C_B = C_B(\alpha)$ such that $u_m(k) \leq C_B m k^{-\alpha-1}$ for all $m \geq 1$ and $k \geq 2m\mu$.*

Proof. Here C depends only on α and may change from line to line. Let $r = 4 + 4/(\alpha - 1)$ and $y = k/r$.

One big jump. By the mean value theorem $\mathbb{P}(\xi = \ell) \leq \alpha \ell^{-\alpha-1}$. So for each i the probability $\mathbb{P}(\tau_m = k, \xi_i \geq y) = \sum_{\ell \geq y} \mathbb{P}(\xi = \ell) \mathbb{P}(\tau_m - \xi_i = k - \ell)$ is at most $\alpha r^{\alpha+1} k^{-\alpha-1}$, and the m indices give at most $\alpha r^{\alpha+1} m k^{-\alpha-1}$.

No big jump. We may take $y \geq 1$, since otherwise $\max_i \xi_i < y$ is impossible. Let $x_0 > 0$ and $\lambda = s/y$ with $s \geq 0$. Markov's inequality and $e^x \leq 1 + x + \frac{x^2}{2} e^x$ for $x \geq 0$ give

$$\mathbb{P}(\tau_m \geq x_0, \max_{i \leq m} \xi_i < y) \leq \exp\left(-\frac{s x_0}{y} + m\left(\frac{s}{y} \mathbb{E}[\xi; \xi < y] + \frac{s^2 e^s}{2y^2} \mathbb{E}[\xi^2; \xi < y]\right)\right). \tag{6}$$

We take $x_0 = k$ and use $\mathbb{E}[\xi; \xi < y] \leq \mu$, $\mathbb{E}[\xi^2; \xi < y] \leq \alpha \sum_{\ell < y} \ell^{1-\alpha} \leq C y^{2-\alpha}$ and $k - m\mu \geq k/2$. The exponent is then at most $-sr/2 + Cr^\alpha s^2 e^s m/k^\alpha$. With $s = \frac{1}{2} \log(k^\alpha/m)$, which is positive since $m \leq k/2 < k^\alpha$, we have $e^s m/k^\alpha = e^{-s}$ and $s^2 e^{-s} \leq 1$. So the bound is $C(m/k^\alpha)^{r/4}$. As $r/4 = 1 + 1/(\alpha - 1)$ and $m \leq k$, this is at most $C(m/k^\alpha)(k^{1-\alpha})^{1/(\alpha-1)} = C m k^{-\alpha-1}$. $\square$

Lemma C.2. *Let D be any a.s. finite first run. Then*

$$\sup_k \left| \frac{a_m}{\mu} q_m(k) - g(z_{m,k}) \right|, \quad \sup_k |a_m u_m^D(k) - g(z_{m,k})|, \quad \sup_k \left| \frac{a_m}{\mu} q_{m+1}^D(k) - g(z_{m,k}) \right|$$

tend to 0 as $m \rightarrow \infty$. For $m \geq 2$, $\sup_k q_m(k) \leq \mu \bar{u}/a_{m-1}$ and $\sup_k u_m^D(k) \leq \bar{u}/a_{m-1}$.

Proof. By (5), q_m and u_m^D are mixtures of shifts of u_{m-1} with total weights μ and 1. This gives the bounds for $m \geq 2$.

Fix $J \geq 1$ and $Z > 0$, put $\sigma_J = \sum_{j>J} \bar{F}(j)$ and write $z = z_{m,k}$. In the sum for $q_m(k)$ the terms with $j > J$ add up to at most $\bar{u}\sigma_J/a_{m-1}$. For $j \leq J$ we have $|a_{m-1}u_{m-1}(k-j) - g(z_{m-1,k-j})| \leq \delta_{m-1}$ and $z_{m-1,k-j} = z a_m/a_{m-1} + (\mu - j)/a_{m-1}$. If $|z| \leq Z$, then $|z_{m-1,k-j} - z| \leq \eta_m = Z(a_m/a_{m-1} - 1) + (J + \mu)/a_{m-1}$. If $|z| > Z$ and $(J + \mu)/a_{m-1} \leq Z/2$, then $g(z)$ and $g(z_{m-1,k-j})$ both lie in $[0, \bar{g}(Z/2)]$. Since $\mu - \sum_{j \leq J} \bar{F}(j) = \sigma_J$, we get, uniformly in k for large m ,

$$|a_{m-1}q_m(k) - \mu g(z)| \leq \mu(\delta_{m-1} + \omega(\eta_m) + \bar{g}(Z/2)) + (\bar{u} + \|g\|_\infty)\sigma_J.$$

We let $m \rightarrow \infty$, then $J, Z \rightarrow \infty$, and use $a_m/a_{m-1} \rightarrow 1$. This gives the first limit. The second follows in the same way from the sum for u_m^D , with the weights $\mathbb{P}(D = d)$ and with $\mathbb{P}(D > J)$ in place of σ_J . The third follows from the sum for q_{m+1}^D , with $a_m u_m^D$ in place of $a_{m-1} u_{m-1}$, the second limit in place of $\delta_{m-1} \rightarrow 0$, and the shift $z_{m,k-j} - z_{m,k} = -j/a_m$. $\square$

Proof of Theorem 4.4. Put $b_h = (h/\mu)^{1/\alpha}$. By Lemma 4.3(iv) it suffices to show that $C_N \sim 2b_h^{-1} \int g^2$. Note that $b_h = o(h)$, that $a_m \geq b_h/8$ for $m \geq h/(8\mu)$ and $a_m \geq b_h/2$ for $m \geq h/(2\mu)$, and that $a_m/b_h \rightarrow 1$ uniformly in m with $|m\mu - h| \leq 2Mb_h$, for each fixed M . By Proposition 4.2, $C_N = \sum_{m \geq 1} [u_m(h)q_{m+1}^D(h) + u_m^D(h)q_m(h)]$. Fix $M > 1$, put $m_\pm = (h \pm Mb_h)/\mu$, and let $\Sigma_1, \dots, \Sigma_4$ be the parts of this sum over the regions $\mathcal{R}_1 = \{m \leq h/(4\mu)\}$, $\mathcal{R}_2 = \{h/(4\mu) < m < m_-\}$, $\mathcal{R}_3 = \{m_- \leq m \leq m_+\}$ and $\mathcal{R}_4 = \{m > m_+\}$.

Region $\mathcal{R}_1$. Here $h \geq 4m\mu$, so $u_m(h) \leq C_B m h^{-\alpha-1} \leq C_B h^{-\alpha}/(4\mu)$ by Lemma C.1. Also $q_1(h) = h^{-\alpha}$. For $m \geq 2$ we split the sum for $q_m(h)$ at $j = h/2$. The terms with $j \geq h/2$ add up to at most $(h/2)^{-\alpha}$. For $j < h/2$ we have $h - j > h/2 \geq 2\mu(m-1)$, so the other terms add up to at most $\mu C_B (m-1)(h/2)^{-\alpha-1} \leq 2^{\alpha-1} C_B h^{-\alpha}$. Since $\sum_m q_{m+1}^D(h) \leq 1$ and $\sum_m u_m^D(h) \leq 1$, we get $\Sigma_1 = O(h^{-\alpha})$. This is $o(1/b_h)$, since $1/\alpha < 1 < \alpha$.

Region $\mathcal{R}_2$. For large h we have $m-1 \geq h/(8\mu)$ here, so $u_m(h), u_m^D(h) \leq \bar{u}/a_{m-1} \leq 8\bar{u}/b_h$ by Lemmas 4.3(iii) and C.2. Let $m' = \lfloor m_- \rfloor$. Then $\sum_{m \in \mathcal{R}_2} q_m(h) \leq \mathbb{P}(\tau_{m'} \geq h)$ and $\sum_{m \in \mathcal{R}_2} q_{m+1}^D(h) \leq \mathbb{P}(D + \tau'_{m'} \geq h)$, which is at most $\mathbb{P}(D \geq b_h) + \mathbb{P}(\tau_{m'} \geq h - b_h)$. Since $m'\mu \leq h - Mb_h$ and $a_{m'} \leq b_h$, the event $\{\tau_{m'} \geq h - b_h\}$ implies $(\tau_{m'} - m'\mu)/a_{m'} \geq M - 1$. As D is a.s. finite and $(\tau_{m'} - m'\mu)/a_{m'} \Rightarrow G$, we get $\limsup_h b_h \Sigma_2 \leq 16\bar{u} \mathbb{P}(G \geq M - 1)$.

Region $\mathcal{R}_4$. For large h we have $m-1 \geq h/(2\mu)$ here, so $u_m(h), u_m^D(h) \leq 2\bar{u}/b_h$. Let $k = \lfloor m_+ \rfloor$. Then $\sum_{m \in \mathcal{R}_4} q_m(h) = \mathbb{P}(\tau_k < h)$ and $\sum_{m \in \mathcal{R}_4} q_{m+1}^D(h) = \mathbb{P}(D + \tau'_k < h) \leq \mathbb{P}(\tau_k < h)$. Now $(h - k\mu)/a_k \leq (\mu - Mb_h)/a_k \rightarrow -M$, and the law of G is continuous, so $\limsup_h b_h \Sigma_4 \leq 4\bar{u} \mathbb{P}(G \leq -M)$.

Region $\mathcal{R}_3$. Here $m \rightarrow \infty$ and $a_m = b_h(1 + o(1))$ uniformly. By Lemma C.2 and $\delta_m \rightarrow 0$, uniformly in $m \in \mathcal{R}_3$,

$$u_m(h)q_{m+1}^D(h) + u_m^D(h)q_m(h) = \frac{2\mu}{a_m^2} (g(z_{m,h})^2 + o(1)).$$

Put $y_m = (h - m\mu)/b_h$. Then $|z_{m,h} - y_m| \leq M|b_h/a_m - 1| \rightarrow 0$, so $g(z_{m,h})^2 = g(y_m)^2 + o(1)$. The y_m with $m \in \mathcal{R}_3$ are the points of an arithmetic progression with step μ/b_h that lie in $[-M, M]$.

Hence

$$b_h \Sigma_3 = 2(1 + o(1)) \frac{\mu}{b_h} \sum_{m \in \mathcal{R}_3} (g(y_m)^2 + o(1)) \rightarrow 2 \int_{-M}^M g^2.$$

Adding the 4 regions,

$$\limsup_{h \rightarrow \infty} \left| b_h C_N - 2 \int g^2 \right| \leq 2 \int_{|z| > M} g^2 + 16\bar{u}(\mathbb{P}(G \geq M - 1) + \mathbb{P}(G \leq -M)),$$

and letting $M \rightarrow \infty$ gives $C_N \sim 2b_h^{-1} \int g^2$. □

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